22.

Algebra

Type - 1

1. Find the numbers if the arithmetic mean and the geometric mean of the two numbers are 7 and $2\sqrt{10}$ respectively.

(a) 5, 4 (b) 2, 20
(c) 4,10 (d) 8,5

RRB RPF Constable – 17/01/2019 (Shift-III)

Ans : (c) Let two number be a and b

Arithmetic mean of both numbers $= \frac{a+b}{2}$

and geometric mean $= \sqrt{ab}$

According to the question,

$\frac{a+b}{2} = 7$

a + b = 14 ....(i)

and $\sqrt{ab} = 2\sqrt{10}$

ab = 40 ...(ii)

Solving of equation (i) and (ii)

$a + \frac{40}{a} = 14$

$\frac{a^2+40}{a} = 14$

$a^2 + 40 = 14a$

$a^2 - 14a + 40 = 0$

$a^2 - 10a - 4a + 40 = 0$

a (a – 10)–4 (a – 10) = 0

(a – 10) (a – 4) = 0

a = 10 or 4

a = 10

b = 4

So numbers are 4 and 10.

2. Find the three numbers in the arithmetic series whose sum is 15 and the product is 105.

(a) 1, 5, 9 or 9, 5, 1 (b) 3, 5, 7 or 7, 5, 3
(c) 3, 8, 7 or 7, 8, 3 (d) 3, 5, 8 or 8, 5, 3

RRB RPF SI – 05/01/2019 (Shift-III)

Ans : (b) From option (b),

3, 5, 7

Succession 5-3=2

7-5=2

from the question, 3+5+7 = 15 (total sum)

$3 \times 7 \times 5 = 105$ (multiple)

3. How many three digit numbers are divisible by 9.

(a) 100 (b) 95
(c) 90 (d) 105

RRB RPF Constable – 19/01/2019 (Shift-II)

Ans : (a) divisible numbers from 9

108, 117, ----------999

first term (a) = 108

last term (l) = 999

common difference (d) = 117-108

= 9

$\because$ l = a + (n – 1) d

999 = 108 + (n – 1) × 9

999 – 108 = (n – 1)9

891 = (n – 1) ×9

99 = (n – 1)

n = 100

So total three digit numbers to be divided from 9 = 100

4. 16, 32, 64, 128, .. 11th term of this series will be

(a) 16348 (b) 16384
(c) 16834 (d) 13684

RRB Group-D – 15/11/2018 (Shift-I)

Ans : (b) 16, 32, 64, 128, ….. 11th term

= 16(1, 2, 4, 8 ----)

n^{th} term of geometric progression

$t_n = ar^{n-1}$

$a = 1, n = 11, r = \frac{4}{2} = 2$

$t_{11} = 16(1\times2^{11-1}) = 16 \times 1024 = 16384$

5. 10th term of arithmetic series $\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}$ is ____

(a) $11\sqrt{2}$ (b) $10\sqrt{2}$
(c) 12 (d) $19\sqrt{2}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (d) Following arithmetic series is-

$\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}$.....

Tenth term = T_{10} = ?

First term $(a) = \sqrt{2}$, common difference $(d) = 2\sqrt{2}$

n = 10 (number of terms)

Tenth term $= T_{10} = \sqrt{2} + (10-1)2\sqrt{2}$

$[\because T_n = a + (n-1)d]$

$T_{10} = \sqrt{2} + 9 \times 2\sqrt{2}$

$T_{10} = \sqrt{2} + 18\sqrt{2}$

$\boxed{T_{10} = 19\sqrt{2}}$

6. The sum of the first 2007 terms of series 1, 2, 3, 4, 1, 2, 3, 4, will be_____

(a) 5016 (b) 5107
(c) 5020 (d) 5013

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (a) 1, 2, 3, 4, 1, 2, 3, 4, _____ 2007 term
Total term to taking number of 1, 2, 3, 4

$= \frac{2007}{4} = 501 + 3$ remainder

sum of number 1, 2, 3, 4 = 10
Total sum = 501×10 +1 +2 + 3
= 5010 + 6 = 5016

7. 17, 22, 27, 32, ... 20th term in this series will be
(a) 107 (b) 117
(c) 112 (d) 115

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (c) 17, 22, 27, 32, ..., 20th term = ?
∵ Above mention series is arithmetic series.

$\therefore \boxed{T_n = a + (n-1)d}$

where a= first term =17, d = common difference = 5
number of n terms = 20
T_{20} = 17 + (20–1) 5
= 17 + 19 × 5
= 17 + 95
= 112

8. What will be the 12th term of arithmetic series 142, 148, 154 _____
(a) 210 (b) 200
(c) 208 (d) 300

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (c) $\boxed{T_n = a + (n-1)d}$
142, 148, 154 ----- 12th term
a = 142
d = 148 - 142 = 6
T_{12} = 142 + (12 - 1) 6
= 142 + 66
= 208

9. The ratio of the sum of the first three terms of the geometric series (G.P.) and the sum of the first six terms is 125 : 152. What is the common ratio of G.P.
(a) $\frac{4}{5}$ (b) $\frac{5}{3}$
(c) $\frac{5}{4}$ (d) $\frac{3}{5}$

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (d)

$$\frac{\text{sum of the six terms of geometric series}}{\text{sum of the three terms of geometric series}} = \frac{152}{125}$$

$$\frac{a + ar + ar^2 + ar^3 + ar^4 + ar^5}{a + ar + ar^2} = \frac{152}{125}$$

$$\frac{a(1 + r + r^2 + r^3 + r^4 + r^5)}{a(1 + r + r^2)} = \frac{152}{125}$$

$$1 + r^3 = \frac{152}{125}$$

$$r^3 = \frac{152}{125} - 1$$

$$r^3 = \frac{152 - 125}{125}$$

$$r^3 = \frac{27}{125}$$

$$r = \frac{3}{5}$$

10. Find the 5th term in the given geometric series 3, 6, 12, ……..
(a) 48 (b) 62
(c) 57 (d) 50

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (a) first term (a) = 3, common ratio (r) $= \frac{6}{3} = 2$
n = 5
formula- $t_n = ar^{n-1}$
$t_5 = 3\,(2)^{5-1} = 3 \times 2^4$
$t_5 = 3 \times 16$
So in series 5th term is 48.

11. What is the 23rd term in the given series 5, 11, 17, ...
(a) 137 (b) 140
(c) 135 (d) 139

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (a) : Given,
First term (a) = 5
Common difference (d) = 6
T_{23} =?
T_{23} = a + (n – 1)d
T_{23} = 5 + (23 – 1)6
T_{23} = 5 + (22 × 6)
T_{23} = 5 + 132
T_{23} = 137

12. The sum of the parallel mean and the geometric mean of two positive numbers is equal to the difference of those numbers. Find the ratio of those numbers.
(a) 9 : 1 (b) 2 : 3
(c) 1 : 4 (d) 1 : 12

RRB Group-D – 15/11/2018 (Shift-II)

Ans : (a) Let number be a and b

arithmetic mean of two positive number $= \frac{a+b}{2}$

Geometric mean $= \sqrt{ab}$

$$\frac{a+b}{2} + \sqrt{ab} = b - a$$

$$a + b + 2\sqrt{ab} = 2(b - a)$$

$$\left(\sqrt{a}\right)^2 + \left(\sqrt{b}\right)^2 + 2\sqrt{a.b} = 2\left[\left(\sqrt{b}\right)^2 - \left(\sqrt{a}\right)^2\right]$$

$$\left(\sqrt{a} + \sqrt{b}\right)^2 = 2\left(\sqrt{b} - \sqrt{a}\right)\left(\sqrt{b} + \sqrt{a}\right)$$

$\sqrt{a}+\sqrt{b}=2\sqrt{b}-2\sqrt{a}$

$3\sqrt{a}=\sqrt{b}$

$\frac{\sqrt{a}}{\sqrt{b}}=\frac{1}{3}$

Square of both side

$\frac{a}{b}=\frac{1}{9}$

So ratio of number = 9 : 1

13. How many numbers of three digits are divisible by 8.

(a) 114 (b) 111
(c) 113 (d) 112

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (d) The smallest number of three digit divisible by 8 is 104 and largest number = 992

$T_n = a + (n-1)\,d$

Where a = first term = 104

Last term (l) = 992

d = common difference = 8

n = number of terms = ?

$l = 104 + (n-1) \times 8$

$992 = 104 + 8n - 8$

$8n = 992 - 96$

$8n = 896$

$n = 112$

So total of 3 digit is 112 number which will be divide by 8.

14. Find the geometric mean of the numbers 7, 7^2, 7^3, ... 7^n

(a) $7^{\frac{n+1}{2}}$ (b) $7^{\frac{n-1}{2}}$
(c) $7^{\frac{7}{4}}$ (d) $7^{\frac{4}{7}}$

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (a) 7, 7^2, 7^3.....7^n Geometric mean of numbers

Formula– Geometric mean $=\sqrt{ab}$

$a = 7,\ r = 7$

$\because$ $T_n = ar^{n-1}$

$T_n = 7.7^{n-1}$

$T_n = 7^n$

Geometric mean $=\sqrt{7.7^n}$

$=\sqrt{7^{n+1}} = 7^{\frac{n+1}{2}}$

15. The sum of the arithmetic progression 1+4+7+ ....+ X is 782. What is the value of x.

(a) 70 (b) 61
(c) 64 (d) 67

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (d) Arithmetic progression 1+4+7+.....X

$S_n = 782$

a = 1 (first term)

d= 4-1 = 3 (common difference)

$S_n = \frac{n}{2}\left[2a+(n-1)d\right]$

$782 = \frac{n}{2}\left[2x1+3n-3\right]$

$1564 = 3n^2 - n$

$3n^2-n-1564=0$

$3n^2-69n+68n-1564=0$

$3n(n-23)+68(n-23)=0$

$(3n+68)\ (n-23)=0$

$n-23=0$

$\boxed{n = 23}$

$T_n = a + (n-1)d = 1 + (23-1)\,3 = 1 + 22 \times 3$

n = 67

16. What is the sum of the first 10 numbers in the sequence –3, –8, –13, –18 .....

(a) –260 (b) –250
(c) –245 (d) –255

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (d) given numbers are in arithmetic progression where,

first term (a) = –3

common difference (d) = –5

number of terms (n) = 10

Sum $(S_n) = \frac{n}{2}\left[2a+(n-1)d\right]$

$S_n = \frac{10}{2}\left[2\times(-3)+(10-1)(-5)\right]$

$S_n = 5\left[-6-45\right] = 5\times(-51)$

$S_n = -255$

17. Split 69 into three parts such that they are in A.P. series and the product of their smaller parts is 483.

(a) 19,23,27 (b) 17,23,29
(c) 15,23,31 (d) 21,23,25

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (d) If three part is a–d, a and a+d then-

According to the question,

$a - d + a + a + d = 69$

$3a = 69$

$a = 23$

and $(a-d) \times a = 483$

$(23-d) \times 23 = 483$

$(23-d) = 21$

or, $d = 2$

Now three part

(23 – 2), 23, and (23 + 2)

$\Rightarrow$ it is 21, 23 and 25.

Type - 2

18. Find the HCF of $(a^3 + b^3)$, $(a + b)^2$ and $(a^2 - b^2)$.

(a) (a + b) (b) (a – b)
(c) (a + b) (a – b) (d) $(a^3 + b^3)\ (a^2 - b^2)$

RRB JE - 31/05/2019 (Shift-III)

Ans. (a) $(a^3 + b^3) = (a + b)\ (a^2 + b^2 - ab)$

$(a + b)^2 = (a + b)\ (a + b)$

and

$(a^2 - b^2) = (a + b)\ (a - b)$

So, HCF = (a + b)

19. Find the LCM of $(a^3 - b^3)$, $(a^2 - b^2)$, $(a - b)$.

(a) $(a - b)$ (b) $(a^3 - b^3)(a + b)$
(c) $(a^3 - b^3)$ (d) $(a^3 - b^3)(a^2 - b^2)$

RRB JE - 27/06/2019 (Shift-I)

Ans : (b) $(a^3 - b^3)$, $(a^2 - b^2)$, $(a - b)$
$(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$
$(a^2 - b^2) = (a - b)(a + b)$
$(a - b) = (a - b)$
LCM $= (a - b)(a + b)(a^2 + b^2 + ab)$
$= (a^3 - b^3)(a + b)$

20. Find HCF of $6xy^2z$, $8x^2y^3z^2$, $12x^3y3^3z^3$

(a) $2xyz$ (b) $4xy^2z$
(c) $3xy^2z$ (d) $2xy^2z$

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) $6xy^2z = 2 \times 3 \times x \times y \times y \times z$
$8x^2 \times y^3 \times z^2 = 2 \times 2 \times 2 \times x \times x \times y \times y \times y \times z \times z$
$12 \times x^3 \times y^3 \times z^3 = 2 \times 2 \times 3 \times x \times x \times x \times y \times y \times y \times z \times z \times z$
HCF $= 2xy^2z$

21. Find HCF of $15x^2+8x-12$, $3x^2 + x - 2$, $3x^2 - 2x$, $9x^2 - 12x + 4$

(a) $x-4$ (b) $x-2$
(c) $3x-4$ (d) $3x-2$

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (d) $15x^2 + 8x - 12$
$= (5x + 6)(3x-2)$
$3x^2 + x - 2$
$= (3x-2)(x+1)$
$3x^2 - 2x = x(3x-2)$
$9x^2 - 12x + 4$
$= (3x-2)(3x-2)$
HCF $= (3x-2)$

22. Find LCM of $2x^3-16$, $3(x^2+3x-10)$, $x^3+2x^2 - 8x$

(a) $6x(x+2)(x+4)(x+5)(x^2+2x+4)$
(b) $6x(x-2)(x+4)(x+5)(x^2+2x+4)$
(c) $6(x-2)(x+4)(x+5)(x^2+2x+4)$
(d) $6x(x-3)(x+4)(x+5)(x^2+2x+4)$

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (b) $2x^3 - 16 = 2(x^3 - 8)$
$= 2(x-2)(x^2 + 4 + 2x)$
$3(x^2 + 3x - 10) = 3(x^2 + 5x - 2x - 10)$
$= 3[x(x+5) - 2(x+5)]$
$= 3(x-2)(x+5)$
$x^3 + 2x^2 - 8x = (x-2)(x^2 + 4x)$
$= x(x-2)(x+4)$
LCM $= 3 \times 2 \times x \times (x-2)(x+4)(x+5)(x^2+4+2x)$
$= 6x(x-2)(x+4)(x+5)(x^2+4+2x)$

Type - 3

23. If there is no solution of the equation $4x+3y+5 = 0$ and $6x-ky-7= 0$, then the value of k will be-

(a) -4.5 (b) -8
(c) 8 (d) 4.5

RRB RPF SI – 12/01/2019 (Shift-III)

Ans : (a)
Given equation is $4x+3y+5=0$ and $6x-ky-7=0$
$a_1x+b_1y+c_1= 0$ and $a_2x+b_2y+c_2=0$ if–

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{4}{6} = \frac{3}{-k} \neq \frac{5}{-7}$$

$$\Rightarrow \frac{4}{6} = \frac{3}{-k} \Rightarrow -4k = 6 \times 3$$

$$\Rightarrow k = \frac{18}{-4}$$

$$\Rightarrow \boxed{k = -4.5}$$

24. Find the value of x and y by solving the following equations: $9x + 3y + 12 = 0$; $18x + 6y + 24 = 0$

(a) $x = 4; y = -16$
(b) $x = 2; y = 10$
(c) $x = 1; y = 7$
(d) not unique solution but infinite solution

RRB JE - 25/05/2019 (Shift-II)

Ans : (d) $9x + 3y + 12 = 0$
compare to equation of $a_1x + b_1y + c_1 = 0$
$a_1 = 9, b_1 = 3, c_1 = 12$
$18x + 6y + 24 = 0$
compare to equation of $a_2x + b_2y + c_2 = 0$
$a_2 = 18, b_2 = 6, c_2 = 24$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{9}{18} = \frac{3}{6} = \frac{12}{24}$$

$$\frac{1}{2} = \frac{1}{2} = \frac{1}{2}$$

So no unique solution but infinite solutions.

25. For which value of p, will have only one solution of the following equation.
$2x + 3y = -5$ and $2x + py = 2$

(a) 3 is the only one solution of p
(b) p has many solution
(c) p has other solution except 3
(d) 2 is the only one solution of p

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (c) For unique solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\frac{2}{2} \neq \frac{3}{p}$$

$$p \neq 3$$

Hence p has any solution except 3

26. If there is no solution of the equations $14x + 8y + 5 = 0$ and $21x - ky - 7 = 0$ is possible, then the value of k will be:

(a) -16 (b) 12
(c) 8 (d) -12

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (d) $14x + 8y + 5 = 0$ — (1)

$21x - ky - 7 = 0$ — (2)

equation (1) and (2) solution will not be possible if-

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{14}{21} = \frac{8}{-k}$$

$$\frac{2}{3} = \frac{8}{-k} \Rightarrow \boxed{k = -12}$$

So k = –12 for value solution will not be possible.

27. If there is no solution of $6x^2 + 2kx + k = 0$, then find the value of k:

(a) $0 < k < 6$ (b) $k > -6$

(c) $k > 6$ (d) $k < 6$

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (a) $6x^2 + 2kx + k = 0$ has no solution

then $\boxed{K = ?}$

put K = 1 in equation

$6x^2 + 2x + 1 = 0$

So there is no solution

on put K = 1 to K = 6

$6x^2 + 12x + 6 = 0$

$6x^2 + 6x + 6x + 6 = 0$

$6x(x + 1) + 6(x + 1) = 0$

$\boxed{x = -1,}$

So there is a solution on equation K = 6

Therefore, it is clear that the equation from K=1 to K=5. There will be no solution.

i.e. $\boxed{0 < K < 6}$

28. What is the value of k for which the equation $16x - 12y + 9 = 0$ and $12x + ky - 11 = 0$ has no solution

(a) –16 (b) 16

(c) –9 (d) 9

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (c) When no solution is there then,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

equation $16x - 12y + 9 = 0$ ...(i)

$12x + ky - 11 = 0$ ...(ii)

from equation (i) and (ii) when there is no solution on comparison then,

$$\frac{16}{12} = \frac{-12}{K} \neq \frac{9}{-11}$$

$$\frac{16}{12} = \frac{-12}{K}$$

$16K = -144$

$$K = \frac{-144}{16}$$

$K = -9$

29. If there is no solution of $20x + 5y + 11 = 0$ and $50x - ky - 9 = 0$, then find the value of k.

(a) 12.5 (b) -12.5

(c) 18 (d) -18

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (b) :

$20x + 5y + 11 = 0$ .....(i)

$50x - ky - 9 = 0$ .....(ii)

$a_1 = 20$ $b_1 = 5$ $c_1 = 11$

$a_2 = 50$ $b_2 = -k$ $c_2 = -9$

There is no solution of 2 linear equation if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

So taking $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

$$\frac{20}{50} = \frac{5}{-k}$$

$$k = -\frac{50 \times 5}{20}$$

$k = -12.5$

30. If no solution of the equations $4x + 3y + 5 = 0$ and $10x - ky - 7 = 0$ is possible, then what will be the value of k?

(a) -8 (b) 7.5

(c) 8 (d) -7.5

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (d) : There is no solution of equation in that the condition

from formula - $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$$\frac{4}{10} = \frac{3}{-k}$$

$-2k = 15$

$\boxed{k = -7.5}$

31. If $3x^2 + kx + k = 0$ has no solution, then the value of k will satisfy :

(a) $k > 12$ (b) $k < 12$

(c) $k > -12$ (d) $0 < k < 12$

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (d) : If there is no solution of $3x^2 + kx + k = 0$ then the origin of the given equation will given equation will be imaginary.

or $b^2 - 4ac < 0$

$\Rightarrow$ $k^2 - 4 \times 3k < 0$

$\Rightarrow$ $k(k - 12) < 0$

$\Rightarrow$ $k - 12 < 0, k < 0$

$k < 12$

If $k < 0$ will be then $k^2 - 12k > 0$

So intended relationship will be = $0 < k < 12$

32. If the equations $6x - 5y + 11 = 0$ and $15x + ky - 9 = 0$ have no solution, then the value of *k* is :

(a) – 18 (b) 12.5

(c) – 12.5 (d) 18

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (c) When equation's no solution then

condition $= \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

As per question,

equation $6x - 5y + 11 = 0$

$15x + ky - 9 = 0$

Then $\frac{6}{15} = \frac{-5}{k} \neq \frac{11}{-9}$

$\frac{2}{5} = \frac{-5}{k}$, $k = \frac{-25}{2}$

$\boxed{k = -12.5}$

33. If $x^2 + kx + k = 0$ has no solution, then the value of k will satisfy:

(a) $k > 4$ (b) $0 < k < 4$

(c) $k < 4$ (d) $k > -4$

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (b) $\because$ there is no solution $x^2 + kx + k = 0$

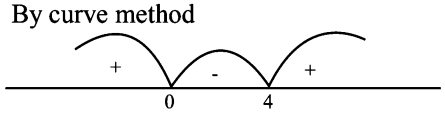
$\therefore$ Discriminant (D) < 0

$b^2 - 4ac < 0$

$k^2 - 4k < 0$

$k(k - 4) < 0$

By curve method



so $0 < k < 4$

34. The pair of linear equations $3x + y = 1$ and $px + 2y = 5$ has no finite solution if :

(a) $0 < p < 6$ (b) $P = 6$

(c) $P = 0$ (d) $p \geq 6$

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (b) Linear equation will not be solution for $3x+y = 1$ and $px+2y = 5$

$\frac{3}{p} = \frac{1}{2} \neq \frac{1}{5} \left\{ \because \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{C_1}{C_2} \right\}$

$\Rightarrow p = 6$

35. The number of solutions of the pair of linear equations $x + 2y - 8 = 0$ and $2x + 4y = 16$ is

(a) 0 (b) 1

(c) infinite (d) 2

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (c)

(I) Unique solution will be equation of $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

(II) No solution of equation $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(III) Infinite solution must be equation of $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

equation $x + 2y - 8 = 0$, $2x + 4y - 16 = 0$

then $\frac{1}{2} = \frac{2}{4} = \frac{-8}{-16}$

$\boxed{\frac{1}{2} = \frac{1}{2} = \frac{1}{2}}$ So infinite solution will be equation.

Type - 4

36. If $(x+1/x) = 3$ then find the value of $(x^3+1/x^3) \div (x^2+1/x^2)$

(a) 18/5 (b) 26/3

(c) 18/7 (d) 54/5

RRB RPF SI – 10/01/2019 (Shift-III)

Ans : (c) $x + \frac{1}{x} = 3$

square of both side

$\left(x + \frac{1}{x}\right)^2 = (3)^2$

$x^2 + \frac{1}{x^2} + 2 \times x \times \frac{1}{x} = 9$

$x^2 + \frac{1}{x^2} = 7$

$x + \frac{1}{x} = 3$

cube of both side

$\left(x + \frac{1}{x}\right)^3 = (3)^3$

$x^3 + \frac{1}{x^3} + 3 \times x \times \frac{1}{x}\left(x + \frac{1}{x}\right) = 27$

$x^3 + \frac{1}{x^3} + 3 \times 3 = 27$

$x^3 + \frac{1}{x^3} = 27 - 9$

$x^3 + \frac{1}{x^3} = 18$

$\frac{x^3 + \frac{1}{x^3}}{x^2 + \frac{1}{x^2}} = \frac{18}{7}$

37. If $a + \frac{1}{a} = -30$ then find the value of $a^3 + \frac{1}{a^3}$

(a) 27090 (b) 26910

(c) -26910 (d) -27090

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (c)

$a + \frac{1}{a} = -30 ----(1)$

$a^3 + \frac{1}{a^3} = ?$

cube of equation (i)

$\left(a + \frac{1}{a}\right)^3 = (-30)^3$

$a^3 + \frac{1}{a^3} + 3\left(a + \frac{1}{a}\right) = -27000$

$$a^3+\frac{1}{a^3}+3(-30)=-27000$$

$$a^3+\frac{1}{a^3}=-27000+90$$

$$\boxed{a^3+\frac{1}{a^3}=-26910}$$

38. If $X^{2n}+\frac{1}{x^{2n}}=k$, then find the value of $x^n-\frac{1}{x^n}$

(a) $k+2$ (b) $k-2$

(c) $\sqrt{k-2}$ (d) $\sqrt{k+2}$

RRB JE - 24/05/2019 (Shift-I)

Ans : (c)

$$\because x^{2n}+\frac{1}{x^{2n}}=k$$

$$\left(x^n-\frac{1}{x^n}\right)^2=x^{2n}+\frac{1}{x^{2n}}-2\times x^n\times\frac{1}{x^n}$$

$$\left(x^n-\frac{1}{x^n}\right)^2=k-2$$

$$\therefore \left(x^n-\frac{1}{x^n}\right)=\sqrt{k-2}$$

39. If $X-\frac{1}{X}=3$ then find the value of $X^4+\frac{1}{X^4}$

(a) 129 (b) 119

(c) 14 (d) 123

RRB JE - 24/05/2019 (Shift-III)

Ans : (b)

$$X-\frac{1}{X}=3$$

$$X^2+\frac{1}{X^2}-2=9$$

$$X^2+\frac{1}{X^2}=11$$

Again square of both side

$$X^4+\frac{1}{X^4}+2=121$$

$$X^4+\frac{1}{X^4}=119$$

40. If $x^3+y^3=9$ and $x+y=3$ then find the value of x^2+y^2

(a) 25 (b) 6

(c) 3 (d) 5

RRB JE - 26/05/2019 (Shift-II)

Ans : (d) Given - $x^3+y^3=9$, $x+y=3$

According to the question,

$(x+y)^3=x^3+y^3+3xy(x+y)$

$(3)^3=9+3xy\times 3$

$27-9=9xy$

$9xy=18$

$\boxed{xy=2}$

$(x+y)^2=x^2+y^2+2xy$

$(3)^2=x^2+y^2+2\times 2$

$9-4=x^2+y^2$

$\boxed{x^2+y^2=5}$

41. If $a-\frac{1}{a}=7$ then $a^2+\frac{1}{a^2}=?$

(a) 52 (b) 50

(c) 49 (d) 51

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d) Given–

$$a-\frac{1}{a}=7$$

Square of both side,

$$a^2+\frac{1}{a^2}-2\times a\times\frac{1}{a}=49$$

$$a^2+\frac{1}{a^2}-2=49$$

$$a^2+\frac{1}{a^2}=49+2$$

$$a^2+\frac{1}{a^2}=51$$

42. If a is positive and $a^2+\frac{1}{a^2}=7$ then $a^3+\frac{1}{a^3}=?$

(a) 18 (b) $3\sqrt{7}$

(c) $7\sqrt{7}$ (d) 21

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a) $a^2+\frac{1}{a^2}=7$ .....(i)

adding 2 in both sides of equation (i),

or $a^2+\frac{1}{a^2}+2\times a\times\frac{1}{a}=7+2$

$$\left(a+\frac{1}{a}\right)^2=(3)^2$$

$$a+\frac{1}{a}=3 \quad .....(ii)$$

Taking cube on both sides of equation (ii)

$$a^3+\frac{1}{a^3}+3a\times\frac{1}{a}\left(a+\frac{1}{a}\right)=27$$

$$a^3+\frac{1}{a^3}+3\times 3=27 \qquad \left(\therefore a+\frac{1}{a}=3\right)$$

$$a^3+\frac{1}{a^3}=27-9$$

$$\boxed{a^3+\frac{1}{a^3}=18}$$

43. If $a+\frac{1}{a}=5$ then find the value of $a^3+\frac{1}{a^3}$?

(a) 140 (b) 110

(c) 120 (d) 130

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (b)

$$a+\frac{1}{a}=5$$

$$a^3+\frac{1}{a^3}=?$$

Now $a+\frac{1}{a}=5$

cube of both sides

$$a^3+\frac{1}{a^3}+3a\times\frac{1}{a}\left(a+\frac{1}{a}\right)=125$$

$$a^3+\frac{1}{a^3}+3\times5=125$$

$$a^3+\frac{1}{a^3}=125-15$$

$$a^3+\frac{1}{a^3}=110$$

44. If $\left(a-\frac{1}{a}\right)=6$ then $\left(a^4+\frac{1}{a^4}\right)=?$

(a) 34 (b) 1444
(c) 38 (d) 1442

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (d) Given by

$$\left(a-\frac{1}{a}\right)=6$$

Square of both side

$$a^2+\frac{1}{a^2}-2=36, \qquad a^2+\frac{1}{a^2}=38$$

Square again

$$a^4+\frac{1}{a^4}+2=1444$$

$$a^4+\frac{1}{a^4}=1442$$

45. If $x^2+\frac{2x}{5}+\frac{1}{25}=0$ then $\left(x-\frac{2}{3}\right)^2=?$

(a) $\frac{1}{3}$ (b) $\frac{169}{225}$
(c) $\frac{44}{169}$ (d) $\frac{225}{256}$

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (b) : $x^2+\frac{2x}{5}+\frac{1}{25}=0$ then $\left(x-\frac{2}{3}\right)^2=?$

$$x^2+\frac{2x}{5}+\frac{1}{25}=0$$

$$25x^2+10x+1=0$$

$$25x^2+5x+5x+1=0$$

$$5x(5x+1)+1(5x+1)=0$$

$$(5x+1)(5x+1)=0$$

$$5x+1=0$$

so $x=-\frac{1}{5}$

Hence $\left(x-\frac{2}{3}\right)^2=\left(-\frac{1}{5}-\frac{2}{3}\right)^2$

$$=\left(-\frac{13}{15}\right)^2=\frac{169}{225}$$

46. If $a-\frac{1}{a}=7$, then $a^2+\frac{1}{a^2}=?$

(a) 51 (b) 53
(c) 52 (d) 54

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (a) Given, $a-\frac{1}{a}=7$

$$\Rightarrow\left(a-\frac{1}{a}\right)^2=(7)^2, \text{ \{square of both side\}}$$

$$\Rightarrow a^2+\frac{1}{a^2}-2\times a\times\frac{1}{a}=49$$

$$\Rightarrow a^2+\frac{1}{a^2}-2=49$$

$$a^2+\frac{1}{a^2}=51$$

47. If $a+\frac{1}{a}=8$, then $a^3+\frac{1}{a^3}=?$

(a) 500 (b) 488
(c) 536 (d) 512

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (b) $a+\frac{1}{a}=8$

cube of both side

$$\left(a+\frac{1}{a}\right)^3=(8)^3$$

$$a^3+\frac{1}{a^3}+3\times a\times\frac{1}{a}\left(a+\frac{1}{a}\right)=512$$

$$a^3+\frac{1}{a^3}+3\times8=512$$

$$a^3+\frac{1}{a^3}=512-24$$

$$a^3+\frac{1}{a^3}=488$$

48. If $a+\frac{1}{a}=-3$ then find the value of $a^6+\frac{1}{a^6}$

(a) 36 (b) 322
(c) 729 (d) 18

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (b)

$$a + \frac{1}{a} = -3$$

square of both side

$$\left(a + \frac{1}{a}\right)^2 = (-3)^2$$

$$a^2 + \frac{1}{a^2} + 2 = 9$$

$$a^2 + \frac{1}{a^2} = 7\text{---------(I)}$$

cube of both side

$$\left(a^2 + \frac{1}{a^2}\right)^3 = (7)^3$$

$$a^6 + \frac{1}{a^6} + 3a^2 \times \frac{1}{a^2}\left(a^2 + \frac{1}{a^2}\right) = 343 \text{ ...from equation (I)}$$

$$a^6 + \frac{1}{a^6} = 343 - 21 = 322$$

49. If $x + y = 9, x^2 + y^2 = 41$, then find the value of $x^3 + y^3$

(a) 189 (b) 249
(c) 289 (d) 100

RRB Group-D – 15/11/2018 (Shift-III)

Ans. : (a) Given by-

$x + y = 9$
$x^2 + y^2 = 41$
$x^3 + y^3 = ?$
$(x + y)^2 = (9)^2$
$x^2 + y^2 + 2xy = 81$
$41 + 2xy = 81$
$xy = 20$
$(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
$(9)^3 = x^3 + y^3 + 3 \times 20 \times 9$
$x^3 + y^3 = 729 - 540$
$x^3 + y^3 = 189$

50. If $\frac{x+1}{x} = 2$, then find the value of $x^2 + \frac{1}{x^2}$

(a) 5 (b) 4
(c) 1 (d) 2

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (d) Given -

$$\frac{x+1}{x} = 2$$

$$\therefore \quad 1 + \frac{1}{x} = 2$$

$$\frac{1}{x} = 2 - 1 = 1$$

$$\boxed{1 = x}$$

so $x^2 + \frac{1}{x^2} = (1)^2 + \frac{1}{(1)^2} = 1 + 1 = 2$

51. If $\frac{x-1}{x} = 3$, then what will be the value of $\frac{x^2+1}{x^2}$

(a) 3 (b) 11
(c) 9 (d) 7

RRB Group-D – 12/10/2018 (Shift-II)

Ans : (*) This question cancelled by RRB

$$\frac{x-1}{x} = 3 \Rightarrow 1 - \frac{1}{x} = 3 \Rightarrow x = -\frac{1}{2}$$

so, $\frac{x^2+1}{x^2} = 1 + \frac{1}{x^2} = 1 + \frac{1}{\left(-\frac{1}{2}\right)^2}$

It will be 1+4= 5 (which answer is not in options)

52. If $a - \frac{1}{a} = 10$, then what will be the value of $a^2 + \frac{1}{a^2}$

(a) 98 (b) 102
(c) 100 (d) 99

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (b) : given–

$$a - \frac{1}{a} = 10$$

square of both side,

$$\left(a - \frac{1}{a}\right)^2 = (10)^2$$

$$\Rightarrow \quad a^2 + \frac{1}{a^2} - 2 = 100$$

$$\Rightarrow \quad a^2 + \frac{1}{a^2} = 100 + 2$$

$$\Rightarrow \quad a^2 + \frac{1}{a^2} = 102$$

53. If (a–1/a) = 3/4 then find the value of (a^3–1/a^3)

(a) 164/31 (b) 171/64
(c) 171/32 (d) 164/37

RRB NTPC 19.04.2016 Shift : 1

Ans : (b) $a - \frac{1}{a} = \frac{3}{4}$

cube of both side

$$\left(a - \frac{1}{a}\right)^3 = \frac{27}{64}$$

$$a^3 - \frac{1}{a^3} - 3\left(a - \frac{1}{a}\right) = \frac{27}{64}$$

$$a^3 - \frac{1}{a^3} - 3 \times \frac{3}{4} = \frac{27}{64}$$

$$a^3 - \frac{1}{a^3} = \frac{27}{64} + \frac{9}{4} = \frac{171}{64}$$

54. If $(x^2 + 1/x^2) = 6$ then find the value of $(10x - 10/x)$

(a) +/–15 (b) +/–20
(c) +/–30 (d) +/–40

RRB NTPC 19.04.2016 Shift : 1

Ans : (b) – $x^2 + \frac{1}{x^2} = 6$

$x^2 + \frac{1}{x^2} - 2 = 6 - 2$

$\left(x - \frac{1}{x}\right)^2 = 4$

$x - \frac{1}{x} = \pm 2$

$10x - \frac{10}{x} = \pm 20$

55. If $(x^2+1/x^2)^2 = 324$ then find the value of $(x-1/x)$

(a) 4 (b) 6 (c) 8 (d) 10

RRB NTPC 18.04.2016 Shift : 3

Ans : (a) Given–

$\left(x^2 + \frac{1}{x^2}\right)^2 = 324$

$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{324}$

$\Rightarrow x^2 + \frac{1}{x^2} = 18$

$\Rightarrow \left(x - \frac{1}{x}\right)^2 + 2 = 18$

$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 18 - 2$

$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 16$

$\Rightarrow x - \frac{1}{x} = \sqrt{16}$

$\Rightarrow x - \frac{1}{x} = 4$

56. If $(x + 1/x) = 2$, then find the value of $(x^3 + 1/x^3) \div (x^{18} + 1/x^{18})$.

(a) 2/9 (b) 5
(c) 1 (d) 1/9

RRB NTPC 12.04.2016 Shift : 3

Ans : (c) From the question,

$x + \frac{1}{x} = 2$

$x^2 + 1 = 2x$

$x^2 - 2x + 1 = 0$

$(x - 1)^2 = 0$

$x - 1 = 0$

$x = 1$

Now

$$\frac{x^3 + \frac{1}{x^3}}{x^{18} + \frac{1}{x^{18}}} = \frac{1+1}{1+1} = \frac{2}{2} = 1$$

57. If $a^2 + \frac{1}{a^2} = 3$ then $a^3 + \frac{1}{a^3} = ?$

(a) $3\sqrt{5}$ (b) $2\sqrt{5}$
(c) $2\sqrt{3}$ (d) $3\sqrt{3}$

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (b) $\left(a + \frac{1}{a}\right)^2 = a^2 + \frac{1}{a^2} + 2$

$\left(a + \frac{1}{a}\right)^2 = 3 + 2$ given $a^2 + \frac{1}{a^2} = 3$

$a + \frac{1}{a} = \sqrt{3+2} = \sqrt{5}$

$a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right)$

$= \left(\sqrt{5}\right)^3 - 3 \times \sqrt{5}$

$= 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}$

so $\boxed{a^3 + \frac{1}{a^3} = 2\sqrt{5}}$

58. If $a - \frac{1}{a} = 1$, then $a^2 + \frac{1}{a^2} = ?$

(a) 1 (b) 3
(c) 2 (d) 4

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (b) $\because$ $a^2 + \frac{1}{a^2} = \left(a - \frac{1}{a}\right)^2 + 2$

$a^2 + \frac{1}{a^2} = 1^2 + 2 = 3$

Type - 5

59. What will be the remainder when $27x^3 - 9x^2 + 3x - 8$ is divided by $3x + 2$

(a) –22 (b) +22
(c) +16 (d) –16

RRB RPF SI – 11/01/2019 (Shift-III)

Ans : (a) $27x^3 - 9x^2 + 3x - 8$,

$3x + 2 = 0$

$3x = -2$

then, $x = \frac{-2}{3}$

so $27\left(\frac{-2}{3}\right)^3 - 9\left(\frac{-2}{3}\right)^2 + 3\left(\frac{-2}{3}\right) - 8$

$\Rightarrow 27\left(\frac{-8}{27}\right) - 9\left(\frac{4}{9}\right) + 3\left(\frac{-2}{3}\right) - 8$

$\Rightarrow -8 - 4 - 2 - 8$

$\Rightarrow -22$

60. If $2x^m + x^3 - 3x^2 - 26$ is divided by $x-2$, the remainder 994 is left, find the value of 'm'

(a) 10 (b) 9
(c) 11 (d) 8

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (b)
On division of $2x^m + x^3 - 3x^2 - 26$ from x–2 left remainder 994 or (x–2) a factor of equation is given
so $x - 2 = 0$
x = 2 putting in equation
$2 \times 2^m + 8 - 12 - 26 = 994$
$2 \times 2^m = 1024$
$2^m = 512$
$2^m = 2^9$
$\boxed{m = 9}$

61. On dividing $2x^2 + ax + b$ by $x - 3$, then remainder is 31 and on dividing $x^2 + bx + a$ $x - 3$, the reaminder is 24 then the value of a + b will be equal to-
(a) -7 (b) 23
(c) -23 (d) 7

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (d) : $2x^2 + ax + b$ ..... (i)
$x^2 + bx + a$ ..... (ii)
dividing to both equation from (x–3) left remainder is 31 and 24 respectively.
So a factor of both equation is (x-3)
∴ putting x = 3 in both equation
from equation (i)
$2 \times 9 + a \times 3 + b = 31$
$3a + b = 13$ .....(iii)
from equation (ii),
$9 + 3b + a = 24$
$a + 3b = 15$ ..... (iv)
equation (iv) × 3
$3a + 9b = 45$
subtract in equation (iv) to equation (iii)
$3a + 9b = 45$
$3a + b = 13$
$- \quad - \quad -$
$8b = 32$
$\boxed{b = 4}$
putting the value of b in equation (iii)
$3a + 4 = 13$
$3a = 9$
$\therefore \boxed{a = 3}$
so $\boxed{a + b = 7}$

62. On dividing $4x^6 - 5x^3 - 3$ by $x^3 - 2$, the remainder left is-
(a) 3 (b) 0
(c) 1 (d) 2

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (a) Divide $4x^6 - 5x^3 - 3$ to $x^3 - 2$
$x^3 - 2 = 0$
$x = (2)^{\frac{1}{3}}$
puttin the value of x in equation
$= 4 \times \left[(2)^{\frac{1}{3}}\right]^6 - 5\left[(2)^{\frac{1}{3}}\right]^3 - 3$
$= 4 \times 2^2 - 5 \times 2 - 3 = 16 - 10 - 3 = 3$
so 3 remainder is left.

63. When $x^2 + ax + b$ is divided by $x - 7$ the remainder left is 35 and when $x^2 + bx + a$ is divided by x - 7, the remainder left is 31. Then a + b is equal to-
(a) 3 (b) 4
(c) –3 (d) –4

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (d) divide in $x^2 + ax + b$ from $x - 7$ remainder = 35
or putting $x = 7$
$(7)^2 + 7a + b = 35$
$7a + b = 35 - 49$
$7a + b = -14$ -------(i)
again divide in $x^2 + bx + a$ from $x - 7$ remainder = 31
or putting $x = 7$
$(7)^2 + 7b + a = 31$
$a + 7b = 31 - 49$
$a + 7b = -18$ -------(ii)
adding equation (i) and (ii) –
$7a + b = -14$
$a + 7b = -18$
$8a + 8b = -32$
$8(a + b) = -32$
$\boxed{a + b = -4}$

64. If $2x^2 + ax + b$ is divided by x – 3, leaves a remainder of 35 and when $2x^2 + bx + a$ is divided by x – 3, leaves a remainder of 29, then the value of a + b will be-
(a) -7 (b) -23
(c) 7 (d) 23

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) $2x^2 + ax + b = 35$ ____ (i)
$2x^2 + bx + a = 29$ _____ (ii)
on putting x = 3 in equation (i) and (ii)
$\Rightarrow 18 + 3a + b = 35$
$\Rightarrow 3a + b = 17$ _____ (iii)
$\Rightarrow 18 + 3b + a = 29$
$\Rightarrow 3b + a = 11$ _____ (iv)
adding equation (iii) and (iv)
$4a + 4b = 28$
$a + b = \frac{28}{4} = 7$
$a + b = 7$

65. If the value for k for which $x^2 + 5kx + k^2 + 5$, is completely divisible by x+2 but not divisible by x+3
(a) neither 1 nor 9 (b) both 1 and 9
(c) 1 (d) 9

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (d) : $x^2 + 5kx + k^2 + 5$, from x + 2 is completely divisible
∴ putting x = –2 remainder = 0
$(-2)^2 + 5k(-2) + k^2 + 5 = 0$
$4 - 10k + k^2 + 5 = 0$
$k^2 - 10k + 9 = 0$
$(k - 9)(k - 1) = 0$

$\therefore$ $k = 9, 1$

putting $k = 9$

$= x^2 + 45x + 81 + 5$

$= x^2 + 45x + 86$

$= x(x + 43) + 2(x + 43)$

$= (x + 43)(x + 2)$

putting $k = 1$

$= x^2 + 5x + 1 + 5$

$= x^2 + 5x + 6$

$= (x + 3)(x + 2)$

but from (x+3) is not divisible

$\therefore$ $k = 9$

66. If $2x^2+ax+2b$, is when divided by x - 1, leaves remainder 16 and when $x^2+bx+2a$ is divided by x + 1, leaves remainder -1 then a+b is equal to-

(a) -8 (b) -14

(c) 14 (d) 8

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : In these questions which to divide. out of that, keep the value of x in equation and write a remainder equal to new equation

$2x^2 + ax + 2b$

putting $x–1 = 0$, $x = 1$

$\therefore$ $2(+1)^2 + a \times (1) + 2b = 16$

$2 + a + 2b = 16$

$a + 2b = 14$ _____ (i)

$x^2 + bx + 2a$

$x + 1 = 0$

putting $x = (-1)$

again $(-1)^2 + b(-1) + 2a = -1$

$2a-b = -2$ _____(ii)

equation (i) + equation (ii) × 2

$a + 2b = 14$

$4a - 2b = -4$

$5a = 10$

$\boxed{a = 2}$

Putting the value of a in equation (i)

$2+2b = 14$

$2b = 12$

$\boxed{b = 6}$

$a + b = 2 + 6$

$= 8$

67. If $3x^2 + ax + 4$, is completely divisible by x – 8 then the value of 'a' will be-

(a) –24.5 (b) 25.5

(c) 24.5 (d) –25.5

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (a) putting x = 8 in given equation

$3x^2 + ax + 4 = 0$

$x – 8 = 0, \Rightarrow x = 8$

$\Rightarrow 3 \times 64 + a \times 8 + 4 = 0$

$\Rightarrow 192 + 8a + 4 = 0$

$\Rightarrow 8a = –192–4$

$\Rightarrow 8a = – 196$

$a = \frac{-196}{8} = -24.5$

$a = –24.5$

68. If $4x^3 – 2x^2 + 5x – 8$ is divided by (x–2) then the remainder will be -

(a) 16 (b) 26

(c) 42 (d) 81

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (b) Divident remainder if $4x^3 – 2x^2 + 5x – 8$ from (x–2)

putting x = 2 in expression $4x^3 – 2x^2 + 5x – 8$

Remainder-

$= 4 \times (2)^3 – 2 \times (2)^2 + 5 \times 2 – 8$

$= 4 \times 8 – 2 \times 4 + 10 – 8$

$= 32 – 8 + 2$

$= \boxed{26}$

69. Find the value of 'A' when the polynomial is P (x) = $x^3 + 3x^2 – 2Ax + 3$ Where 'A' is the constant which divided by $x^2 + 1$ the remainder is equal to -5x

(a) 3 (b) -2

(c) 2 (d) -3

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (c) Polynomial P (x) = $x^3+3x^2–2Ax+3$

Remainder on divident = -5x

So remainder theorem $x^2 + 1 = 0$

putting the value of $x^2 = –1$ in polynomial

P (x) = $-x^2x + 3x^2 – 2Ax + 3$

$(-1) \times x + 3\times(-1) – 2Ax + 3 = -5x$

$–x – 3 – 2Ax + 3 = –5x$

$– 2Ax = –4x$

$A = 2$

70. When $3x^2 + 2ax + 4b$ is divided by x + 3, leaves a remainder of 15 and when $2x^2 + 3bx + 5a$ is divided by x-3, leaves a remainder 65 then the value of a+b is.

(a) 9 (b) 6

(c) 7 (d) 11

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (c) : First expression = $3x^2 + 2ax + 4b$

second expression = $2x^2 + 3bx + 5a$

divide to x + 3 from first expression left remainder is 15

$x + 3 = 0$

putting in first expression x = -3

$3 \times (-3)^2 + 2a \times (-3) + 4b = 15$

$27 – 6a + 4b = 15$

$3a – 2b = 6$ .....(i)

divide to x–3 from second expression left remainder is 65

$x – 3 = 0$

x = 3 putting in second expression

$2(3)^2 + 3b (3) + 5a = 65$

$18 + 9b + 5a = 65$

$5a + 9b = 47$ .....(ii)

equation (i) × 9 equation. (ii) × 2

$27a – 18b = 54$

$10a + 18b = 94$

to add

$37a = 148$

$a = 4$

putting the value of a in equation (i)

$3a - 2b = 6$

$3 \times 4 - 2b = 6$

$12 - 6 = 2b$

$6 = 2b$

$b = 3$

so $b = 3$

then $a + b = 4 + 3 = 7$

71. On dividing $4x^6 - 5x^3 - 3$ by $x^3 - 2$, leaves a remainder.

(a) 3 (b) 2

(c) 0 (d) 1

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (a) According to the question,

$$\begin{array}{r|l} & 4x^3+3 \\ \hline x^3-2 & 4x^6-5x^3-3 \\ & \underline{4x^6-8x^3} \\ & \quad -\quad + \\ & \quad 3x^3-3 \\ & \quad \underline{3x^3-6} \\ & \quad -\quad + \\ & \qquad 3 \end{array}$$

So remainder will be 3

72. If $3x^2 + ax - 12$ is completely divisible by x-8 then the value of 'a' is-

(a) -24.5 (b) 22.5

(c) 24.5 (d) -22.5

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (d) $3x^2 + ax - 12$ .....(i)

equation (i) is totally divisible by $x - 8$

$\therefore$ putting $x = 8$,

$3 \times (8)^2 + a \times 8 - 12 = 0$

$3 \times 64 + 8a - 12 = 0$

$8a = 12 - 192$

$8a = -180$

$a = -22.5$

73. If $3x^2 + ax + 7$ is completely divisible by x-1 then the value of 'a' is-

(a) –10.6 (b) –10.5

(c) –10 (d) –3

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (c)

equation $3x^2 + ax + 7$ is divisible by $x - 1$

then $x - 1 = 0$

putting $x = 1$

$\Rightarrow$ $3(1)^2 + a(1) + 7 = 0$

$\Rightarrow$ $3 + a + 7 = 0$

$\boxed{a = -10}$

74. If $4x^3 - 2x^2 + 5x - 8$ is divided by (x–2) then the remainder is-

(a) 43 (b) 16

(c) 26 (d) 25

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (c) $x - 2 = 0$

$x = 2$

putting the value of x in given expression -

Remainder $= 4(2)^3 - 2(2)^2 + 5 \times 2 - 8$

$= 4\times8 - 2\times4 + 5 \times 2 - 8$

$= 32 - 8 + 10 - 8 = 42 - 16 = 26$

So remainder $= 26$

75. $15y^3 - 30y^2 + 12y - 12$ is divided by 3y–6, what will be the remainder.

(a) 6 (b) 36

(c) 30 (d) 12

RRB NTPC 29.04.2016 Shift : 2

Ans : (d) $\because 3y - 6 = 0$

$\therefore 3y = 6$

$\boxed{y = 2}$

Putting the value of y in the expression

$= 15(2)^3 - 30(2)^2 + 12(2) - 12$

$= 15 \times 8 - 30 \times 4 + 24 - 12$

$= 120 - 120 + 12 = 12$

So remainder will be = 12

76. If $x^2 + ax + b$, divided by $x - 3$, then the remaining 22 is obtained and the expression $x^2 + bx + a$, when divided by $x - 3$ then the remaining 24 is obtained. What is the value of a+b?

(a) 23 (b) – 23

(c) – 7 (d) 7

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (d) Left remainder is 22 if divide x^2+ax+b to x-3

$\therefore$ $x^2 + ax + b = 22$ (putting $x = 3$)

$9 + 3a + b = 22$

$3a + b = 13$ ....(i)

$x^2 + bx + a = 24$ (putting $x = 3$)

$9 + 3b + a = 24$

$3b + a = 15$ ....(ii)

from equation (i) + (ii)

$4(a + b) = 28$

$a + b = \frac{28}{4}$

$\boxed{a + b = 7}$

77. If $x^2 + ax + b$, when divided by $x - 4$, left a remainder of 32 and $x^2 + bx + a$, when divided by x–4, left a remainder of 35, then a + b = ?

(a) –7 (b) 23

(c) –23 (d) 7

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (d) $\because$

left remainder is 32 from $x^2 + ax + b \div (x - 4)$

$\therefore$ totally divide (x-4) from $x^2 + ax + b - 32 = 0$

$x - 4 = 0$

putting $x = 4$

$(4)^2 + 4a + b - 32 = 0$

$\Rightarrow$ $16+4a+b-32 = 0$

$\Rightarrow$ $4a+b = 16$ ..... (i)

and left remainder is 35 from $(x^2+bx+a) \div (x-4)$

$\therefore x^2 + bx + a - 35 = 0$, it will be totally divisible from x–4

$\Rightarrow$ $x - 4 = 0$

$\Rightarrow$ putting $x = 4$

$(4)^2 + b \times 4 + a - 35 = 0$

$16 + 4b + a - 35 = 0$
$4b + a - 19 = 0$
$4b+a = 19$ ..... (ii)
Adding equation (i) + (ii)
$5a+5b = 16+19$
$5(a+b) = 35$
$a+b= 7$

78. If $x^2 + ax + b$, divided by x+3 then the remaining – 1 is obtained and the expression $x^2 + bx + a$ when divided by x – 3 then the remaining 39 is obtained. What is the value of a+b?
(a) – 14 (b) – 38
(c) 14 (d) 38
RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (c) $x^2 + ax + b = -1$
putting $x = -3$
$9 - 3a + b = -1$
$b - 3a = -10$ ..... (1)
$x^2 + bx + a = 39$
putting $x = 3$
$9 + 3b + a = 39$
$3b + a = 30$ ..... (2)
solving equation from (1) and (2)
$a = 6, b = 8$
$a + b = 6 + 8 = 14$

Type - 6

79. Factors of $x^2 + 7x + 10$ are.
(a) (x – 5) (x – 2) (b) (x + 5) (x + 2)
(c) (x – 5) (x + 2) (d) (x – 4) (x + 2)
RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (b) factor of $x^2 + 7x + 10$
$= x^2 + 5x + 2x + 10$
$= x (x+5) + 2 (x+5)$
$= (x+5) (x+2)$

80. $x^3+5x^2-2x-24$ has a zero x=2. Find the other zero.
(a) -3, 5 (b) -2, 3
(c) -3, -4 (d) 3, 4
RRB JE - 24/05/2019 (Shift-III)

Ans : (c) (x-2) is a factor of $x^3 + 5x1^2 - 2x - 24$
$x^2 (x -2) + 7x^2 - 2x - 24 = 0$
$x^2 (x - 2) + 7x (x - 2) + 12x -24 = 0$
$x^2 (x - 2) + 7x (x - 2) + 12 (x -2) = 0$
$(x^2 + 7x + 12) (x - 2) = 0$
$(x + 4) (x + 3) (x - 2) = 0$
$x = -4, -3, 2$

81. If (x – 1) and (x + 3), is factor of $x^2 + ax + b$, then find the value of 'a' and 'b' respectively.
(a) 2, 3 (b) 2, – 3
(c) –2, –3 (d) –2, 3
RRB JE - 25/05/2019 (Shift-I)

Ans. (b) equation x^2+ax+b has a factor of (x–1) and (x+3)
$(x - 1) (x + 3) = x^2 + ax + b$
$x^2 + 3x - x - 3 = x^2 + ax + b$
$x^2 + 2x - 3 = x^2 + ax + b$
compare of both sides
$a = 2$ $b = -3$

82. What is the squarroot of $(x^2+4x+4) (x^2+6x+9)$
(a) (2x + 3) (x + 3) (b) (x + 2) (x+3)
(c) (x + 2) (2x + 3) (d) (x+2) (x+4)
RRB Group-D – 31/10/2018 (Shift-II)

Ans : (b) $(x^2 + 4x + 4) (x^2 + 6x + 9)$
$(a^2 + 2ba+b^2) (a^2 + 2.a.b + b^2)$ comprising
$(a+b)^2 (a+b)^2$
$\{(x^2 + 2.2x + (2)^2\} \{(x^2 + 2.3. x + (3)^2\}$
$(x+2)^2 (x+3)^2$
square root = $\boxed{(x+2)(x+3)}$

83. Find the factors of $(x^2 + x - 42)$
(a) (x+14) (x–3) (b) (x+6) (x–7)
(c) (x–6) (x+7) (d) (x–14) (x+3)
RRB NTPC 30.03.2016 Shift : 1

Ans : (c) $x^2 + x - 42$
$= x^2 + 7x - 6x - 42$
$= x(x+ 7) - 6 (x+7)$
$= (x-6) (x+7)$

84. Select the correct factor of $f(x) = 2x^2 - 5x + 2$
(a) x – 2 (b) x – 3
(c) x – 4 (d) x –5
RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (a) $f (x) = 2x^2 - 5x + 2$
$= 2x^2 - 4x - x + 2$
$= 2x (x - 2) - 1 (x - 2)$
$= (2x - 1) (x - 2)$

85. Find the factors of $x^2 + x - 20$
(a) (x + 5) (x – 4) (b) (x + 4) (x – 5)
(c) (x – 2 (x + 10) (d) (x – 2) (x + 5)
RRB NTPC 09.04.2016 Shift : 3

Ans : (a) $x^2 + x - 20$
$= x^2 + 5x - 4x -20$
$= x (x+5) - 4 (x+ 5)$
$= (x + 5) (x - 4)$

86. If the factor of $3x^4 - (a + 2) x^3 - x^2 - 4$ is (x-2), then find the value of 'a'
(a) 5 (b) -1
(c) 3 (d) 4
RRB NTPC 22.04.2016 Shift : 1

Ans : (c) Given -
factor of $3x^4 - (a + 2) x^3 - x^2 - 4 = (x -2)$
$\because$ x -2, is a factor
$\therefore$ $x-2 = 0 \Rightarrow x = 2$
$\Rightarrow 3\times(2)^4 - (a+2)\times(2)^3 - (2)^2 - 4 = 0$
$\Rightarrow 3\times16 - (a+2)\times8 - 4 - 4 = 0$
$\Rightarrow 48 - 8a - 16 - 8 = 0$
$\Rightarrow 24 - 8a = 0$
$\Rightarrow 8a = 24$
$\Rightarrow a = \frac{24}{8}$
$\Rightarrow \boxed{a = 3}$

87. Factorise : $(x^2 - 13x - 48)$
(a) (x - 4) (x + 12) (b) (x + 4) (x - 12)
(c) (x - 16) (x + 3) (d) (x + 16) (x - 3)
RRB NTPC 26.04.2016 Shift : 1

Ans : (c) $x^2 - 13x - 48$
$= x^2 - 16x + 3x - 48$
$= x(x-16) + 3(x-16)$
$= (x-16)(x+3)$

88. Factors of $x^2 - 8x + 12$
(a) $(x-6)(x-2)$ (b) $(x-6)(x+2)$
(c) $(x-4)^2$ (d) $(x+6)(x-2)$
RRB NTPC 27.04.2016 Shift : 1

Ans : (a) $x^2 - 8x + 12$
$= x^2 - 6x - 2x + 12$
$= x(x-6) - 2(x-6)$
$= (x-6)(x-2)$

89. Factors of $x^2 - 6x + 8$
(a) $(x-4)(x-2)$ (b) $(x+4)(x+2)$
(c) $(x+8)(x-2)$ (d) $(x-4)(x+2)$
RRB NTPC 29.04.2016 Shift : 2

Ans : (a) given expression $x^2 - 6x + 8$
$= x^2 - 4x - 2x + 8$
$= x(x-4) - 2(x-4)$
$= (x-4)(x-2)$

90. Factors of $x^2 - 2x - 15$
(a) $(x+5)(x-3)$ (b) $(x-5)(x-3)$
(c) $(x-3)(x-1)$ (d) $(x-5)(x+3)$
RRB NTPC 29.04.2016 Shift : 3

Ans. (d) : from the question
$x^2 - 2x - 15$
$= x^2 - 5x + 3x - 15$
$= x(x-5) + 3(x-5)$
$= (x-5)(x+3)$

91. Factors of $x^2 - 6x + 8$
(a) $(x+4)(x-2)$ (b) $(x-4)(x+2)$
(c) $(x-4)^3$ (d) $(x-4)(x-2)$
RRB NTPC 30.04.2016 Shift : 1

Ans : (d) $x^2 - 6x + 8$
$= x^2 - 4x - 2x + 8$
$= x(x-4) - 2(x-4)$
$= (x-4)(x-2)$

92. Factors of $x^2 + 6x + 8$
(a) $(x+4)(x+2)$ (b) $(x-4)(x+2)$
(c) $(x-)2$ (d) $(x-4)(x-2)$
RRB NTPC 30.04.2016 Shift : 2

Ans : (a) $x^2 + 6x + 8$
$= x^2 + 4x + 2x + 8$
$= x(x+4) + 2(x+4)$
$= (x+4)(x+2)$

Type - 7

93. If there is two real solution of $x^2 - 4x + 4b = 0$, then find the value of 'b'.
(a) $b \geq 1$ (b) $b < 1$
(c) $b = +1, -1$ (d) $b = 0$
RRB JE - 22/05/2019 (Shift-I)

Ans : (b) $x^2 - 4x + 4b = 0$
Discriminant's value must be positive for two real solution of quadratic equations
or
$b^2 - 4ac > 0$
$(-4)^2 - 4 \times 1 \times 4b > 0$
$16 - 16b > 0$
$16 > 16b$
$1 > b$
so $b < 1$

94. If the two roots of the quadratic equation is α andβ, when α+β=8 and α-β=2, then equation is
(a) $x^2 - 8x + 15 = 0$ (b) $x^2 + 8x - 15 = 0$
(c) $x^2 - 8x - 15 = 0$ (d) $x^2 + 8x + 15 = 0$
RRB JE - 23/05/2019 (Shift-II)

Ans : (a)
∵ α and β quadratic equations have two roots.
$\alpha + \beta = 8$ ..... (i)
$\alpha - \beta = 2$ ..... (ii)
$2\alpha = 10$
$\alpha = 5$
from equation (i) –
$5 + \beta = 8$
$\beta = 3$
quadratic equation
$x^2 - (\alpha + \beta)x + \alpha\beta = 0$
$x^2 - 8x + 15 = 0$

95. Find the root of $2x^2 - 15x + 28$
(a) Both negative
(b) Not real
(c) One positive, one negative
(d) Both positive
RRB JE - 27/05/2019 (Shift-I)

Ans : (d)
$2x^2 - 15x + 28 = 0$
$2x^2 - (8+7)x + 28 = 0$
$(2x^2 - 8x) - (7x - 28) = 0$
$2x(x-4) - 7(x-4) = 0$
$(2x-7)(x-4) = 0$
$2x - 7 = 0$
$x = 7/2$
$x - 4 = 0$
$x = 4$
It is clear that both roots is positive.

96. For which value of 'K' are the roots of the equation kx(x-2) + 6 = 0 is similar.
(a) 6 (b) 0, 6
(c) 3, 2 (d) -2
RRB JE - 27/05/2019 (Shift-III)

Ans : (a) $kx(x-2) + 6 = 0$
$kx^2 - 2kx + 6 = 0$
roots of quadratic equations will be equal if value of these discriminant will be zero.
$b^2 - 4ac = 0$
$(-2k)^2 - 4 \times k \times 6 = 0$
$4k^2 - 24k = 0$
$4k(k-6) = 0$
$k = 6$

97. If a and b are the roots of equation $3x^2 - 5x + 2 = 0$ then find the value of (a/b)+(b/a).

(a) 13/9 (b) 13/6
(c) 13/2 (d) 9/13

RRB JE - 28/05/2019 (Shift-II)

Ans : (b) Comparing equation $Ax^2 + Bx + C = 0$ from equation $3x^2 - 5x + 2 = 0$

A = 3, B = -5, C = 2

sum of roots $(a + b) = \frac{-B}{A} = \frac{-(-5)}{3} = \frac{5}{3}$

Multiplication of roots $(ab) = \frac{C}{A} = \frac{2}{3}$

$$\frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab}$$

$$= \frac{a^2 + b^2 + 2ab - 2ab}{ab}$$

$$= \frac{(a+b)^2 - 2ab}{ab}$$

$$= \frac{\left(\frac{5}{3}\right)^2 - 2 \times \frac{2}{3}}{\frac{2}{3}}$$

$$= \frac{\frac{25}{9} - \frac{4}{3}}{\frac{2}{3}}$$

$$= \frac{\frac{25-12}{9}}{\frac{2}{3}}$$

$$= \frac{13}{9} \times \frac{3}{2} = \frac{13}{6}$$

98. For which value of 'K' in equation $x^2 + 2Kx + 4 = 0$, is a real solution?

(a) 0 (b) 2, 0
(c) –2, 0 (d) 2, – 2

RRB JE - 31/05/2019 (Shift-I)

Ans : (d) Real solution is $x^2 + 2Kx + 4 = 0$ for the value of K

$B^2 + 4AC = 0$
$4k^2 - 16 = 0$
$k^2 = 4$
$k = \pm 2$

99. Given $2x^2+19x+45 = 0$ and $2y^2+11y+12 = 0$ then which of the following is true for the roots x, y.

(a) $x \geq y$ (b) $x < y$
(c) $x \leq y$ (d) $x > y$

RRB JE - 31/05/2019 (Shift-III)

Ans. (b) $2x^2 + 19x + 45 = 0$
$2x^2 + 10x + 9x + 45 = 0$
$2x(x + 5) + 9(x + 5) = 0$
$(x + 5)(2x + 9) = 0$
$x = -5, -9/2$

$\because$ $2y^2 + 11y + 12 = 0$
$2y^2 + 8y + 3y + 12 = 0$
$2y(y + 4) + 3(y + 4) = 0$
$(y + 4)(2y + 3) = 0$
$y = -4, -3/2$

It is clear that $y > x$

100. Find the equation whose roots are $(a+\sqrt{b})$ and $(a-\sqrt{b})$.

(a) $x^2 + 2ax - (a^2 - b) = 0$
(b) $x^2 - 2ax + (a^2 - b) = 0$
(c) $x^2 - ax + a^2 - b^2 = 0$
(d) $x^2 + ax + a^2 - b^2 = 0$

RRB JE - 02/06/2019 (Shift-III)

Ans : (b) Equation's roots is $(a+\sqrt{b})$, $(a-\sqrt{b})$

x^2 – (sum of roots) x + multiplication of roots = 0

$x^2 - (a+\sqrt{b} + a-\sqrt{b})x + (a+\sqrt{b})(a-\sqrt{b}) = 0$
$x^2 - 2ax + (a^2 - b) = 0$

101. If α and β are the roots of quadratic equation $(5+\sqrt{2})x^2 - (4+\sqrt{5})x + (8+2\sqrt{5}) = 0$ then the value of $\frac{2\alpha\beta}{\alpha+\beta}$ will be.

(a) 4 (b) 2
(c) 8 (d) 7

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (a) $\alpha + \beta = \frac{-b}{a} = \frac{+(4+\sqrt{5})}{(5+\sqrt{2})}$

$$\alpha\beta = \frac{c}{a} = \frac{8+2\sqrt{5}}{5+\sqrt{2}}$$

$$\frac{2\alpha\beta}{\alpha+\beta} = \frac{2 \times \left(\frac{8+2\sqrt{5}}{5+\sqrt{2}}\right)}{\left(\frac{4+\sqrt{5}}{5+\sqrt{2}}\right)}, = 4 \times \frac{(4+\sqrt{5})}{(5+\sqrt{2})} \times \frac{(5+\sqrt{2})}{(4+\sqrt{5})} = 4$$

102. A root of $x^2 - 12x + 2k = 0$ is x = 4. The other root will be -

(a) x = 6 (b) x = 8
(c) x = –4 (d) x = –8

RRB RPF SI – 06/01/2019 (Shift-I)

Ans. (b) : $x^2 - 12x + 2k = 0$ putting x = 4 in

$(4)^2 - 12 \times 4 + 2k = 0$
$16 - 48 + 2k = 0$
$2k = 32$
$\boxed{k = 16}$

New equation

$x^2 - 12x + 32 = 0$
$x^2 - 12x + 32 = 0$
$x^2 - 8x - 4x + 32 = 0$
$x(x-8) - 4(x-8) = 0$
$(x-8)(x-4) = 0$
$x - 8 = 0 \Rightarrow x = 8$

$\therefore$ 0 then root = 8

103. Solve the following equation to find the value of 'x'. $(x-2)^2 - 36 = 0$; $x \in N$

(a) 4 (b) –8
(c) –4 (d) 8

RRB Group-D – 03/10/2018 (Shift-I)
RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (d)
$(x-2)^2 - 36 = 0$ $x \in N$
$x^2 + 4 - 4x - 36 = 0$
$x^2 - 4x - 32 = 0$
$x^2 - 8x + 4x - 32 = 0$
$x(x-8) + 4(x-8) = 0$
$(x-8)(x+4)$
so x = 8, –4 but according to the question $x \in N$ so value of x will be 8.

104. Two square roots of a quadratic equation is given as $x = \frac{1}{7}$ and $x = \frac{-1}{8}$. The equation can be written as–

(a) $(7x + 1)(8x + 1) = 0$
(b) $(7x - 1)(8x - 1) = 0$
(c) $(7x + 1)(8x - 1) = 0$
(d) $(7x-1)(8x+1) = 0$

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (d)
$x = \frac{1}{7}, \quad x = -\frac{1}{8}$
$\Rightarrow 7x = 1, \quad 8x = -1$
$(7x - 1) = 0, (8x + 1) = 0$
quadratic equations will be–
$\boxed{(7x-1)(8x+1)=0}$

105. Solve- $x : x \in N; (x-4)^2 - 36 = 0$

(a) 2 (b) –2
(c) 10 (d) –10

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (c) $(x-4)^2 - 36 = 0$
$x^2 + 16 - 8x - 36 = 0$
$x^2 - 8x - 20 = 0$
$x^2 - 10x + 2x - 20 = 0$
$x(x-10) + 2(x-10) = 0$
$(x+2)(x-10) = 0$
$x - 10 = 0$
$\boxed{x = 10}$
$x + 2 = 0$
$x = -2$
Always it is taken as positive value.

106. One of the roots of equation $x^2 - 24x + k = 0$ is x = 2 other value will be–

(a) x = 12 (b) x = –22
(c) x = 22 (d) x = –12

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (c) : $x^2 - 24x + k = 0$
one root x = 2 then putting the value of x in equation
$(2)^2 - 24 \times 2 + k = 0$
$4 - 48 + k = 0$
$k - 44 = 0$
$k = 44$
substituting the value of k into the equation-
$x^2 - 24x + 44 = 0$
$x^2 - 22x - 2x + 44 = 0$
$x(x-22) - 2(x-22) = 0$
$(x-2)(x-22) = 0$
x = 2 and x = 22
(given)
so other root = 22

107. If $a + \frac{1}{a} = -6$ then find the value of $a^3 + \frac{1}{a^3}$.

(a) -198 (b) -216
(c) 216 (d) 198

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (a) : $a + \frac{1}{a} = -6$
Cube of both sides
$\left(a + \frac{1}{a}\right)^3 = (-6)^3$
$a^3 + \frac{1}{a^3} + 3 \times a \times \frac{1}{a}\left(a + \frac{1}{a}\right) = (-6)^3$
$a^3 + \frac{1}{a^3} + 3\left(a + \frac{1}{a}\right) = -216$
$a^3 + \frac{1}{a^3} + 3 \times (-6) = -216$
$a^3 + \frac{1}{a^3} = -216 + 18$
$a^3 + \frac{1}{a^3} = -198$

108. If $x^2+1.5kx+4.5k=0$ contains the recurring root, what will be the satisfactory value of k.

(a) only k = 8 (b) k < 0 or k > 8
(c) k = 8 or k = 0 (d) 0 < k < 8

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (c) Quadratic equations-
comparing from $ax^2 + bx + c$
$x^2 + 1.5kx + 4.5k = 0$
a = 1, b = 1.5k, c = 4.5k
$b^2 - 4ac = 0$
$(1.5k)^2 - 4 \times 1 \times 4.5k = 0$
$\frac{225k^2}{100} - 18k = 0$
$225k^2 - 1800k = 0$
$225k^2 = 1800k$
$\boxed{k = 8}$ or k = 0

109. If α and β are o (zero) of the polynomial $x^2 - 5x + 6$. Then find the value of $\frac{\alpha^2 + \beta^2}{\alpha^{-2} + \beta^{-2}}$=?

(a) 63 (b) 62
(c) 36 (d) 26

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (c)
Quadratic equations $ax^2 + bx + c = 0$ of root (zero) is α and β then $\alpha+\beta=-\frac{b}{a}$

and $\alpha.\beta=\frac{c}{a}$

given polynomial $x^2 - 5x + 6$

$\therefore \alpha+\beta = 5$

$\alpha.\beta = 6$

$$\therefore \frac{\alpha^2+\beta^2}{\alpha^{-2}+\beta^{-2}}=\frac{\alpha^2+\beta^2}{\alpha^{\frac{1}{2}}+\beta^{\frac{1}{2}}}=\frac{\alpha^2+\beta^2}{\alpha^2+\beta^2}.\alpha^2\beta^2=\alpha^2\beta^2$$

$$=(\alpha\beta)^2$$

$$=6^2=36$$

110. If $x^2 + 4.5kx + 13.5k = 0$ have recurring roots, then what condition will the value to satisfy.

(a) $0<k<\frac{8}{3}$ (b) $k<0$ or $k>\frac{8}{3}$

(c) $k=\frac{8}{3}$ (d) $k=\frac{8}{3}$ or $k=0$

RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (d) $x^2 + 4.5\,kx + 13.5\,k = 0$

$$x^2+\frac{9}{2}kx+\frac{27}{2}k=0$$

if root is equal/same then–

$b^2 - 4ac = 0$

$$\left(\frac{9}{2}\right)^2 k^2-4\times\frac{27}{2}k=0$$

$$\frac{81}{4}k^2-54k=0$$

$$k\left\{\frac{81}{4}k-54\right\}=0$$

$k = 0,$

or $81k = 54 \times 4$

$$k=\frac{8}{3}$$

111. Two roots of a quadratic equation are $x=\frac{1}{11}$ and $x=\frac{-1}{9}$. The equation can be written as which of the following.

(a) $(11x+1)(9x-1)=0$
(b) $(11x+1)(9x+1)=0$
(c) $(11x-1)(9x+1)=0$
(d) $(11x-1)(9x-1)=0$

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (c) When two roots given α and β

$$\alpha=\frac{1}{11},\ \beta=\frac{-1}{9}$$

quadratic equation written as-

$$(x-\alpha)(x-\beta)=0$$

$$\left(x-\frac{1}{11}\right)\left(x+\frac{1}{9}\right)=0$$

$$\left(\frac{11x-1}{11}\right)\left(\frac{9x+1}{9}\right)=0$$

$\Rightarrow$ $(11x-1)(9x+1)=0$

112. If $3x^2+2kx+k=0$ has no real solution, then what will be correct about the value of k.

(a) $k<0$ or $k>3$ (b) $k>3$
(c) $0<k<3$ (d) $k<0$

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (c) When root is not real

by formula $b^2-4ac<0$

$3x^2+2kx+k=0$

$a=3,\ b=2k,\ c=k$

$4k^2-12k<0$

$4k(k-3)<0$

$0<k<3$

113. If $\alpha \neq \beta$ but $\alpha^2=5\alpha-3, \beta^2=5\beta-3$, then find the equation whose roots are $\frac{\alpha}{\beta},\frac{\beta}{\alpha}$.

(a) $3x^2-19x-3=0$ (b) $3x^2-19x+3=0$
(c) $3x^2+19x-3=0$ (d) $3x^2+19x+3=0$

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (b) Given-

$\alpha \neq \beta$

$\alpha^2=5\alpha-3,$ $\beta^2=5\beta-3$

$\alpha^2-5\alpha+3=0$ $\beta^2-5\beta+3=0$

$$\alpha=\frac{-(-5)\pm\sqrt{(-5)^2-4\times1\times3}}{2}$$

$$\beta=\frac{\left(+5\pm\sqrt{(-5)^2-4\times1\times3}\right)}{2}$$

$$\alpha=\frac{+5+\sqrt{25-12}}{2},\ \beta=\frac{+5-\sqrt{25-12}}{2}$$

$$\boxed{\alpha=\frac{+5+\sqrt{13}}{2}} \qquad \boxed{\beta=\frac{+5-\sqrt{13}}{2}}$$

Formula of quadratic equation -

x^2 – (sum of roots) x + multiplication of roots = 0

$$x^2-\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right)x+\frac{\alpha}{\beta}\times\frac{\beta}{\alpha}=0$$

$$x^2-\left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right)x+1=0$$

$$x^2-\left[\frac{\left(\frac{+5+\sqrt{13}}{2}\right)^2+\left(\frac{+5-\sqrt{13}}{2}\right)^2}{\left(\frac{+5+\sqrt{13}}{2}\right)\left(\frac{+5-\sqrt{13}}{2}\right)}\right]x+1=0$$

$$x^2-\frac{76}{25-13}x+1=0$$

$$x^2-\frac{76}{12}x+1=0$$

$$12x^2-76x+12=0$$

$$\Rightarrow \boxed{3x^2-19x+3=0}$$

114. If $2x^2+x-28<0$, then which of the following specifies all the possible values of 'x'.

(a) $x>\frac{7}{2}$ (b) $-4<x<\frac{7}{2}$

(c) $0<x<13$ (d) $x<-4$

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (b) $2x^2+x-28<0$

$2x^2+8x-7x-28<0$

$2x(x+4)-7(x+4)<0$

$(x+4)(2x-7)<0$

if $x+4<0$

$x>-4$

if $x-7<0$

$x<\frac{7}{2}$

So All possible value of $-4<x<\frac{7}{2}$, x is specified.

115. If α and β are the roots of $2x^2+7x-4=0$, then what will be the equation whose roots are α^2 and β^2.

(a) $4x^2-65x-16=0$

(b) $4x^2+65x-16=0$

(c) $4x^2-65x+16=0$

(d) $4x^2+65x+16=0$

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (c) Equation $2x^2+7x-4=0$ is root of α, β

or $\alpha+\beta=\frac{-b}{a}=\frac{-7}{2}$

$\alpha.\beta=\frac{c}{a}=\frac{-4}{2}=-2$

if root of equation is α^2 and β^2 then

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\frac{49}{4}-2\times(-2)=\frac{65}{4}$

$\alpha^2\beta^2=(\alpha\beta)^2=4$

quadratic equations–

x^2-(sum of root) $x+$(product of root) $=0$

$=x^2-\frac{65}{4}x+4=0$

$4x^2-65x+16=0$

116. If $2x^2-9x-18<0$, then which of the following specifies all the possible values of 'x'.

(a) $x<-\frac{3}{2}x$ (b) $-\frac{3}{2}<x<6$

(c) $x<6$ (d) $0<x<12$

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (b)

$2x^2-9x-18<0$

$\Rightarrow 2x^2-(12-3)x-18<0$

$\Rightarrow 2x^2-12x+3x-18<0$

$\Rightarrow 2x(x-6)+3(x-6)<0$

$\Rightarrow (x-6)(2x+3)<0$

$x-6<0$

Then $\boxed{x<6}$.....(i)

$2x+3<0$

$\boxed{x>\frac{-3}{2}}$.....(ii)

from equation (i) and (ii),

$-\frac{3}{2}<x<6$

117. What is the number of real roots of the equation? $(6-x)^4+(8-x)^4=16$

(a) 6 (b) 2

(c) 4 (d) 0

RRB Group-D – 31/10/2018 (Shift-III)

Ans : (a) Putting x = 6

$(6-6)^4+(8-6)^4=16$

$2^4=16$

$16=16$

Putting = 8,

$(6-8)^4+(8-8)^4=16$

$(-2)^4=16$

So it is clear that two real root of equation will be 6 and 8.

118. If the roots α and β of the equation $x^2-x-1=0$ then what will be the equation whose roots will be $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(a) $x^2+3x-1=0$ (b) $x^2+x-1=0$

(c) $x^2-x+1=0$ (d) $x^2+3x+1=0$

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (a) : root of $x^2-x-1=0$ is α and β then–

$\alpha+\beta=\frac{-b}{a}=-(-1)=1$ ..... (i)

and $\alpha.\beta=\frac{c}{a}=\frac{-1}{1}=-1$, ... (ii)

square of equation (i)

$\alpha^2+\beta^2+2\alpha\beta=1$

$\alpha^2+\beta^2-2=1$

$\alpha^2+\beta^2=3$ [from equation (i)]

that equation whose root are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

x^2 – (sum of roots) x + (multiplication of root)

$$x^2-\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right)x+\frac{\alpha}{\beta}\times\frac{\beta}{\alpha}=0$$

$$x^2-\left(\frac{\alpha^2+\beta^2}{\alpha.\beta}\right)x-1=0$$

$x^2+3x-1=0$

119. If $12x^2-ax+7 = ax^2+9x+3$ has only one (repeated) solution, then the positive integral solution of a is:

(a) 2 (b) 4
(c) 3 (d) 5

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (c) $12x^2-ax+7=ax^2+9x+3$

$x^2(12-a)-x(9+a)+4=0$ .....(i)

equation (i) compare to quadratic equation $ax^2+bx+c=0$
root of equation (i) will be equale (repeated)
if $b^2=4ac$
so taking a=3 from option (c) will be same root (repeated)

120. The value of k, for which the quadratic equation $4x^2+4\sqrt{3}x+k=0$ has equal roots is:

(a) –2 (b) 3
(c) 2 (d) –3

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (b) Same quadratic equation's root will be same if discriminant will be $D=b^2-4ac=0$

here, in $4x^2+4\sqrt{3}x+k=0$

$a=4, b=4\sqrt{3}$ and $c=k$

$(4\sqrt{3})^2-4\times4\times k=0$ {from $=b^2-4ac$}

$48-16k=0$

or $16k=48$

$\boxed{k=3}$

121. One of the roots of the equation $x^2-6x+k=0$ is x = 2. The other root is:

(a) x = 4 (b) x = –1
(c) x = –4 (d) x = 1

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (a) equation $x^2-6x+K=0$

x = 2 satisfies the equation

$2^2-6\times2+K=0$

K = 8

Now equation $x^2-6x+8=0$

(x–2) (x–4) = 0, x =2,x = 4

So second root of equation will be $\boxed{x=4}$

122. If $x^2+kx+k=0$ has repeated roots, then the value of k will satisfy:

(a) k < 0 or k > 4 (b) only k = 4
(c) k = 4 or k = 0 (d) 0 < k < 4

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (c) putting D = 0

$D=b^2-4ac=0$..... (1)

so comparing equation $ax^2+bx+c=0$ from $x^2+kx+k=0$

a = 1 b = k c = k

So putting value in equation (1)

$k^2-4\times1\times k=0$

$k^2-4k=0$

$k(k-4)=0$

$\boxed{k=0}$

$k-4=0$

then, $\boxed{k=4}$

So given option (c) is correct.

123. If $x^2+kx+k=0$ has two distinct real solutions, then the value of k will satisfy:

(a) 0 < k < 4
(b) k < 0 only
(c) k > 4 only
(d) k < 0 or k > 4

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (d) Root is real and unequal

then $b^2-4ac>0$

According to the question,

$k^2-4k>0$

$k(k-4)>0$

$\boxed{k>4,\ k<0}$

124. The two roots of an quadratic equation are given an $x=\frac{1}{2}$ and $x=\frac{-1}{3}$. The equation can be written as:

(a) (2x–1) (3x–1) = 0 (b) (2x+1) (3x–1) = 0
(c) (2x+1) (3x+1) = 0 (d) (2x–1) (3x+1) = 0

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (d) Given root– $x=\frac{1}{2},\ x=-\frac{1}{3}$

equation,

$$\left(x-\frac{1}{2}\right)\left(x-\left(\frac{-1}{3}\right)\right)=0$$

$$\left(x-\frac{1}{2}\right)\left(x+\frac{1}{3}\right)=0$$

$\Rightarrow$ $(2x-1)(3x+1)=0$

125. One quadratic roots of the equation $x^2-4x+k=0$ is x = 3. The other root is:

(a) x = – 1 (b) x = – 4
(c) x = 1 (d) x = 4

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (c) Given-

$x^2-4x+k=0$ ....(i)

x = 3 ....(ii)

From equation (i) and (ii)

$9-4\times3+k=0$

k = 3

So

$x^2 - 4x + 3 = 0$
$x^2 - 3x - x + 3 = 0$
$x(x-3) - 1(x-3) = 0$
$(x-3)(x-1) = 0$
$x = 3, 1$
other root is 1.

Type - 8

126. What is the sum of the values of x satisfying $x^{\frac{2}{3}} + x^{\frac{1}{3}} = 2$.

(a) 7 (b) -3
(c) 3 (d) -7

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (d)

$x^{\frac{2}{3}} + x^{\frac{1}{3}} = 2$

cube of both side

$\left(x^{\frac{2}{3}} + x^{\frac{1}{3}}\right)^3 = (2)^3$

$x^2 + x + 3 \times 2.x = 8 \Rightarrow x^2 + 7x - 8 = 0$

$\Rightarrow (x+8)(x-1) = 0$

$\boxed{x = -8, 1}$

So sum of x value's $= -8 + 1$
$= -7$

127. The numbers of solution of equation $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$ is.

(a) 1 (b) 3
(c) 4 (d) 2

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (d) $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$

$x^{\frac{2}{3}} + x^{\frac{1}{3}} = 2 ----(i)$

cube's both sides of equations

$x^2 + x + 3\left(x^{\frac{2}{3}} + x^{\frac{1}{3}}\right).x = 8$

$x^2 + x + 3x \times 2 = 8$

$x^2 + 7x - 8 = 0$

This is a quadratic equations. So number of solutions will be 2.

128. If $x < 5$ then, the sign of the value of the expression $-5x^2 + 2x + 7$ will be

(a) Can not be said (b) Non negative
(c) Negative (d) Positive

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (a) Since the value of x is less than 5 but not indicated that the value of x will be positive that negative. Hence the value of the given expression can not be determind.

Type - 9

129. Select the value which in the following expression will replace '?' If a+b+c = 0 then $a^3 + b^3 + c^3 = ? \times abc$

(a) 1 (b) 4
(c) 3 (d) 2

RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (c) We know that

$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ ---(i)

Given-

$a + b + c = 0$
$a^3 + b^3 + c^3 = ? \times abc$

$\frac{a^3 + b^3 + c^3}{abc} = ?$ ---(ii)

Putting a + b + c = 0 in equation (i)

$a^3 + b^3 + c^3 - 3abc = 0$
$a^3 + b^3 + c^3 = 3abc$ ---(iii)

from equation (ii) and (iii)

$\frac{3abc}{abc} = 3$

130. If a(a+b+c) = 45; b (a + b + c) = 75 and c (a+ b + c) = 105 then find the value of $a^2 + b^2 + c^2$.

(a) 83 (b) 225
(c) 625 (d) 90

RRB JE - 25/05/2019 (Shift-III)

Ans : (a) $a(a + b + c) = 45$ .....(i)
$b(a + b + c) = 75$ .....(ii)
$c(a + b + c) = 105$ ..... (iii)

from equation (i) and (ii),

$\frac{a}{b} = \frac{3}{5}$ .....(iv)

from equation (ii) and (iii),

$\frac{b}{c} = \frac{5}{7}$ ..... (v)

from equation (iv) and (v),

$a : b : c = 3 : 5 : 7$

from eqation (i),

$3k(3 + 5 + 7)k = 45$
$45k^2 = 45$
$k^2 = 1$
$k = 1$

$\therefore$ $a^2 + b^2 + c^2 = 3^2 + 5^2 + 7^2$
$= 9 + 25 + 49$
$= 83$

131. If $x^{1/3} + y^{1/3} - z^{1/3} = 0$, then find the value of $(x + y - z)^3 + 27xyz$.

(a) $x^3 + y^3 - z^3$ (b) $-z^3$
(c) 1 (d) 0

RRB JE - 28/05/2019 (Shift-II)

Ans : (d) $x^{\frac{1}{3}} + y^{\frac{1}{3}} - z^{\frac{1}{3}} = 0$

$x^{\frac{1}{3}} + y^{\frac{1}{3}} = z^{\frac{1}{3}}$.....(i)

cube of both side

$$\left(x^{\frac{1}{3}}+y^{\frac{1}{3}}\right)^3=\left(z^{\frac{1}{3}}\right)^3$$

$$\left(x^{\frac{1}{3}}\right)^3+\left(y^{\frac{1}{3}}\right)^3+3x^{\frac{1}{3}}y^{\frac{1}{3}}\left(x^{\frac{1}{3}}+y^{\frac{1}{3}}\right)=z$$

from equation (i)

$$x+y+3x^{\frac{1}{3}}y^{\frac{1}{3}}z^{\frac{1}{3}}=z \qquad \left(\because x^{\frac{1}{3}}+y^{\frac{1}{3}}=z^{\frac{1}{3}}\right)$$

$$x+y-z=-3(xyz)^{\frac{1}{3}}$$

again cube of both side,

$$(x+y-z)^3=\left[-3(xyz)^{\frac{1}{3}}\right]^3$$

$(x + y - z)^3 = -27\ xyz$

$(x + y - z)^3 + 27\ xyz = 0$

132. If a + b + c = 0, then find the value of $(b + c)^2/bc + (c + a)^2/ca + (a + b)^2/ab$.

(a) $a^2 + b^2 + c^2$ (b) $2(a + b + c)^2$

(c) 8 abc (d) 3

RRB JE - 31/05/2019 (Shift-III)

Ans : (d) $a + b + c = 0$

$a + b = -c$

$(a + b)^2 = c^2$

$$\frac{(a+b)^2}{ab}=\frac{c^2}{ab} \qquad \text{----(i)}$$

Similarly,

$$\frac{(b+c)^2}{bc}=\frac{a^2}{bc} \qquad \text{----(ii)}$$

$$\frac{(c+a)^2}{ca}=\frac{b^2}{ca} \qquad \text{----(iii)}$$

from equation (i) + equation (ii) + equation (iii)-

$$\frac{(a+b)^2}{ab}+\frac{(b+c)^2}{bc}+\frac{(c+a)^2}{ca}=\frac{c^2}{ab}+\frac{a^2}{bc}+\frac{b^2}{ca}$$

$$=\frac{c^3+a^3+b^3}{abc}$$

$$=\frac{3abc}{abc}$$

$$\frac{(a+b)^2}{ab}+\frac{(b+c)^2}{bc}+\frac{(c+a)^2}{ca}=3$$

133. If a + b + c = 2s, then find the value of $(s - a)^3 + (s - b)^3 + 3(s - a)(s - b)c$.

(a) 2s (b) c^2

(c) c^3 (d) ac

RRB JE - 01/06/2019 (Shift-II)

Ans : (c) $a + b + c = 2s$ -----(i)

$(s-a)^3+(s-b)^3+3(s-a)(s-b) = (s-a+s-b)^3$

$= (2s-a-b)^3$

$= (a+b+c-a-b)^3$

$= c^3$

134. $$\frac{(a-b)^3+(b-c)^3+(c-a)^3}{3(a-b)(b-c)(c-a)}=?$$

What is the value of above expression?

(a) 1 (b) 4

(c) 0 (d) 2

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (a)

$$\frac{(a-b)^3+(b-c)^3+(c-a)^3}{3(a-b)(b-c)(c-a)}=?$$

Let $a - b = A$

$b - c = B$

$c - a = C$

so $A + B + C = a - b + b - c + c - a$

$= 0$

$\therefore A^3 + B^3 + C^3 = 3ABC$

or

formula- $(a-b)^3 + (b-c)^3 + (c-a)^3 = 3(a-b)(b-c)(c-a)$

so

$$=\frac{3(a-b)(b-c)(c-a)}{3(a-b)(b-c)(c-a)}$$

$= 1$

135. If a+b+c = 16 and ab+bc+ca = 78. Find the value of $a^3 + b^3 + c^3 - 3abc$.

(a) 218 (b) 352

(c) 320 (d) 220

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (b) :

$a+b+c=16$

$ab+bc+ca=78$

Find the value of $a^3+b^3+c^3-3abc=?$

$(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$

$(16)^2=a^2+b^2+c^2+2ab+2bc+2ca$

$(16)^2-2(ab+bc+ca)=a^2+b^2+c^2$

$256-2\times 78=a^2+b^2+c^2$

$256-156=a^2+b^2+c^2$

$100=a^2+b^2+c^2$

$a^3+b^3+c^3-3abc=(a+b+c)\{(a^2+b^2+c^2)-(ab+bc+ca)\}$

$(a+b+c)\{(a^2+b^2+c^2)-(ab+bc+ca)\}$

$16\times(100-78)$

$\Rightarrow 16\times 22$

$\Rightarrow 352$

136. If a+b+c = 9 and $a^2+b^2+c^2 = 29$ then find the value of $a^3 + b^3 + c^3 - 3abc$.

(a) 9 (b) 27

(c) 3 (d) 81

RRB Group-D – 31/10/2018 (Shift-I)

Ans : (b) If $a + b + c = 9$ $a^2 + b^2 + c^2 = 29$

then $a^3 + b^3 + c^3 - 3abc = ?$

formula = $(a + b+c)^2 = a^2 + b^2 + c^2+2 (ab+bc+ca)$

$(9)^2 = 29 + 2 (ab + bc + ca)$

$81-29 = 2(ab+bc+ca)$
$26 = (ab+bc+ca)$
formula- $a^3+b^3+c^3-3abc =$
$(a+b+c)[(a^2+b^2+c^2)-(ab+bc+ca)]$
$= 9(29-26)$
$= 9 \times 3$
$\boxed{a^3+b^3+c^3-3abc = 27}$

137. If 2x (x + y + z) = 250, 2y (x + y + z) = 100, 2z (x + y + z) = 100 then find the value of (3x + 6y + 15z).
(a) 110 (b) 95
(c) 85 (d) 69
RRB NTPC 19.04.2016 Shift : 1

Ans : (b) $2x(x+y+z) = 250$ …… (1)
$2y(x+y+z) = 100$ …… (2)
$2z(x+y+z) = 100$ …….. (3)
Adding the equation of (1), (2) and (3)
$(x+y+z)(2x+2y+2z) = 450$
$2(x+y+z)^2 = 450$
$(x+y+z)^2 = 225$
$x+y+z = 15$
$\therefore$ from equation (1), $x = \frac{250}{30} = \frac{25}{3}$
from equation (2), $y = \frac{100}{30} = \frac{10}{3}$
from equation (3), $z = \frac{100}{30} = \frac{10}{3}$
$\therefore 3x + 6y + 15z$
$= 3\times\frac{25}{3} + 6\times\frac{10}{3} + 15\times\frac{10}{3} = 25 + 20 + 50 = 95$

138. If a = 5, b = 4, c = 8 then find the value of $(a^3 + b^3 + c^3 - 3abc)/(ab + bc + ca - a^2 - b^2 - c^2)$.
(a) 15 (b) 17
(c) –17 (d) –15
RRB NTPC 19.04.2016 Shift : 2

Ans : (c) Given-
$a = 5, b = 4, c = 8$
$$\frac{a^3+b^3+c^3-3abc}{(ab+bc+ca-a^2-b^2-c^2)} = \frac{(a+b+c)(a^2+b^2+c^2-ab-bc-ca)}{-(a^2+b^2+c^2-ab-bc-ca)}$$
$= -(a+b+c) = -(5+4+8) = -17$

139. If $a^3 + b^3 + c^3 - 3abc = 0$ then find the value of $(a^2/bc + b^2/ac - 3)$.
(a) $-c^2/ab$ (b) $-c^2/bc$
(c) $-c^3/ba$ (d) $-c/a$
RRB NTPC 19.04.2016 Shift : 2

Ans : (a) Given-
$a^3 + b^3 + c^3 - 3abc = 0$ ……(i)
$\therefore \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} - 3 = 0$ (divide to abc in both side)
$\frac{a^2}{bc} + \frac{b^2}{ca} - 3 = -\frac{c^2}{ab}$
$= \boxed{\frac{-c^2}{ab}}$

140. For real a, b and c if $a^2 + b^2 + c^2 = ab + bc + ca$, then find the value of $(a + b + c)^2$.
(a) $9a^2$ (b) $81a^2$
(c) $27a^2$ (d) $243a^2$
RRB NTPC 18.04.2016 Shift : 1

Ans : (a) from–
$a^2 + b^2 + c^2 = ab + bc + ca$
similarly $a^2 = ab$
$b^2 = bc$
$c^2 = ca$
so $a = b = c$ ----------(1)
so from the quesiton,
$\therefore (a+b+c)^2 = (a+a+a)^2$
$= 9a^2$

141. Given is a = 128, b = 130 and c = 132, find the value of a + b + c - 3abc
(a) 5000 (b) 4680
(c) 4280 (d) 4890
RRB NTPC 16.04.2016 Shift : 2

Ans : (b) Given-
$a = 128, b = 130, c = 132$
$a^3 + b^3 + c^3 - 3abc$
$= \frac{1}{2}(a+b+c)\{(a-b)^2 + (b-c)^2 + (c-a)^2\}$
$= \frac{1}{2}(128+130+132)$
$\{(128-130)^2 + (130-132)^2 + (130-132)^2 + (132-128)^2\}$
$= \frac{1}{2}\times 390\{(-2)^2 + (-2)^2 + (4)^2\}$
$= 195 \times 24 = 4680$

142. If $a^2 + b^2 + c^2 + 3 = 2(a + b + c)$ then the value of (a + b + c).
(a) 2 (b) 5
(c) 4 (d) 3
RRB NTPC 12.04.2016 Shift : 1

Ans : (d) from the question,
$a^2 + b^2 + c^2 + 3 = 2(a+b+c)$
$\Rightarrow a^2 + b^2 + c^2 + 3 - 2(a+b+c) = 0$
$\Rightarrow (a^2 - 2a + 1) + (b^2 - 2b + 1) + (c^2 - 2c + 1) = 0$
$\Rightarrow (a-1)^2 + (b-1)^2 + (c-1)^2 = 0$
$\Rightarrow \therefore a - 1 = 0$ | $b - 1 = 0$ | $c - 1 = 0$
$a = 1$ | $b = 1$ | $c = 1$
$\therefore a + b + c = 1 + 1 + 1$
$= 3$

143. If (a + b + c) = 6 and $a^2 + b^2 + c^2 = 14$ then (ab + bc + ca) = ?
(a) 22 (b) 11
(c) 33 (d) 44
RRB NTPC 06.04.2016 Shift : 2

Ans : (b) $(a+b+c) = 6$, $a^2+b^2+c^2 = 14$, $(ab+bc+ca) = ?$
$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$
$(6)^2 = 14 + 2(ab+bc+ca)$
$36 - 14 = 2(ab+bc+ca)$
$22 = 2(ab+bc+ca)$
$ab + bc + ca = 11$

144. If $w = -2, x = 3, y = 0$ and $z = -\frac{1}{2}$ then find the value of $2(w^2 + x^2 + y^2)$

(a) 26 (b) -26
(c) 25 (d) 28

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) Given by,

$$w = -2,\ x = 3,\ y = 0,\ z = -\frac{1}{2}$$

from the question,

$$2(w^2 + x^2 + y^2) = ?$$

$$\Rightarrow \quad 2\left\{(-2)^2 + 3^2 + 0^2\right\}$$

$$\Rightarrow \quad 2(4 + 9 + 0)$$

$$\Rightarrow \quad 2 \times 13 = 26$$

145. Given is $w = -2,\ x = 3,\ y = 0$ and $z = -\frac{1}{2}$, find the value of $x^2(z + wy)$:

(a) –4.5 (b) 4.5
(c) 2 (d) –2

RRB NTPC 26.04.2016 Shift : 3

Ans : (a) $w = -2$, x=3, y=0, z=$-\frac{1}{2}$

$$x^2(z + wy) = (3)^2\left(-\frac{1}{2} + (-2 \times 0)\right)$$

$$= 9\left(\frac{-1}{2} + 0\right)$$

$$= 9 \times \frac{-1}{2} = \frac{-9}{2} = -4.5$$

146. If a + b + c = 0, then $(a^3 + b^3 + c^3)^2$ = ?

(a) $3a^2b^2c^2$ (b) $9a^2b^2c^2$
(c) 9abc (d) 27abc

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (b) According to the formula–

$a^3+b^3+c^3-3abc = (a + b + c)(a^2+b^2+c^2-ab-bc-ca)$

$\because$ Given– $(a + b + c) = 0$

$\therefore$ $a^3 + b^3 + c^3 - 3abc = 0\ (a^2 + b^2 + c^2 - ab - bc - ca)$

$a^3 + b^3 + c^3 - 3abc = 0$

$a^3 + b^3 + c^3 = 3abc$

Square of both side–

$(a^3 + b^3 + c^3)^2 = (3abc)^2$

$$\boxed{\left(a^3 + b^3 + c^3\right)^2 = 9a^2b^2c^2}$$

Type - 9

147. If $2^x - 2^{x-1} = 8$ then find the value of $2x^2+4x+3$.

(a) 41 (b) 20
(c) 21 (d) 51

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans : (d) $2^x - 2^{x-1} = 8$

$\Rightarrow 2^x(1 - 2^{-1}) = 2^3$

$\Rightarrow 2^x\left(1 - \frac{1}{2}\right) = 2^3$

$\Rightarrow 2^x\left(\frac{1}{2}\right) = 2^3$

Multiply by 2 in both side-

$2^x = 2^4$

$\Rightarrow x = 4$

Put, x = 4 from the equation- $(2x^2+4x+3)$

$2x^2 + 4x + 3 = 2 \times (4)^2 + 4 \times 4 + 3$

$= 2\times16 + 16 + 3 = 32 + 19 = 51$

148. If $a^{x+y} = a^6$ and x is 2 more than y find x.

(a) 1 (b) 2
(c) 3 (d) 4

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (d) $a^{x+y} = a^6 \Rightarrow x + y = 6$.....(i)

from the question $x = y + 2 \Rightarrow$ $\boxed{y = x - 2}$

from the equation (i)-

$x + (x - 2) = 6$

$2x = 8 \Rightarrow x = 4$

149. If $\frac{p}{b-c} = \frac{q}{c-a} = \frac{r}{a-b}$ then find the value of p+q+r .

(a) -1 (b) 1
(c) 2 (d) 0

RRB JE - 28/06/2019 (Shift-III)

Ans. (d) $\frac{p}{b-c} = \frac{q}{c-a} = \frac{r}{a-b} = k$ (suppose)

p = (b–c)k = bk–ck

q = (c–a)k = ck–ak

r = (a–b)k = ak–bk

p+q+r = bk+ck+ak–ck–ak–bk = 0

150. IF y = 5 then find the value of $5y\sqrt{y^3 - y^2}$.

(a) 500 (b) 250
(c) 50 (d) $50\sqrt{2}$

RRB JE - 26/05/2019 (Shift-II)

Ans : (b) Given–

$y = 5, \quad 5y\sqrt{y^3 - y^2} = ?$

$= 5y\sqrt{y^3 - y^2} \quad = 5 \times 5\sqrt{(5)^3 - (5)^2}$

$= 25\sqrt{125 - 25} \quad = 25\sqrt{100}$

$= 25 \times 10 \quad = 250$

151. If 3x – y = 5 then find the value of $8^x/2^y$.

(a) 32 (b) 256
(c) 64 (d) 16

RRB JE - 27/05/2019 (Shift-II)

Ans : (a) 3x – y = 5 ..... (i)

$\because \quad \frac{8^x}{2^y} = \frac{2^{3x}}{2^y}$

$= 2^{3x-y}$ (from equation (i))

$= 2^5$

$= 32$

152. Evaluate $\sqrt{72+\sqrt{72+\sqrt{72+\sqrt{72+\sqrt{.....}}}}}$?

(a) 9 (b) 8
(c) $\sqrt{72}$ (d) 8.8

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (a) $\sqrt{72+\sqrt{72+\sqrt{72+\sqrt{72+.....}}}} = x$.....(i)

Square of both side,

$72+\sqrt{72+\sqrt{72+\sqrt{72+.....}}} = x^2$

from the equation (i)

$72 + x = x^2$

$x^2 - x - 72 = 0$

$x^2 - 9x + 8x - 72 = 0$

$x(x-9)+8(x-9)=0$

$(x+8)(x-9)=0$

$x-9=0, \quad x+8=0$

so $x = 9 \quad x = -8$ (invalid)

153. Find the value $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{....\infty}}}}}$?

(a) $\sqrt{2}$ (b) 2
(c) $2\sqrt{2}$ (d) 4

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (b) $\sqrt{2\sqrt{2\sqrt{2\sqrt{2.....\infty}}}}$

Let $x = \sqrt{2\sqrt{2\sqrt{2\sqrt{2.....\infty}}}}$ .....(1)

Square of equation (1)-

$x^2 = 2\sqrt{2\sqrt{2\sqrt{2\sqrt{2.....\infty}}}}$

$x^2=2x$

$x^2-2x=0$

$x(x-2) = 0$

$x = 0, 2 \quad x\neq0$

So, $\boxed{x=2}$

154. Solve:

$(a^{-1} + b^{-1}) \div (a^{-2} - b^{-2})$

(a) $\frac{ab}{b-a}$ (b) $\frac{a+b}{ab}$
(c) $\frac{ab}{a-b}$ (d) $\frac{a}{b}$

RRB NTPC 18.01.2017 Shift : 2

Ans : (a) $(a^{-1} + b^{-1}) \div (a^{-2} - b^{-2})$

$= \frac{\frac{1}{a}+\frac{1}{b}}{\frac{1}{a^2}-\frac{1}{b^2}} = \frac{\frac{a+b}{ab}}{\frac{b^2-a^2}{a^2b^2}} = \frac{a+b}{ab} \times \frac{a^2b^2}{(a+b)(b-a)}$

$\Rightarrow \boxed{\frac{ab}{b-a}}$

155. If Y^2+Y=12 then Y=?

(a) 2 (b) -1
(c) 4 (d) 3

RRB NTPC 31.03.2016 Shift : 3

Ans : (d) $Y^2 + Y = 12$

$\Rightarrow Y^2 + Y - 12 = 0$

$\Rightarrow Y^2 + 4Y - 3Y - 12 = 0$

$\Rightarrow Y(Y+4) - 3(Y+4) = 0$

$\Rightarrow (Y-3)(Y+4) = 0$

$Y = 3, -4$

So $\boxed{Y=3}$

156. Simplify $(2x)^2 - (2y)^2 - (4x)^2$

(a) $-12x^2 + 4y^2$ (b) $12x^2 - 4y^2$
(c) $-12x^2 - 4y^2$ (d) $12x^2 + 4y^2$

RRB NTPC 17.01.2017 Shift-1

Ans : (c) $(2x)^2 - (2y)^2 - (4x)^2$

$=4x^2 - 4y^2 - 16x^2$

$= -12x^2 - 4y^2$

or $-(12x^2 + 4y^2)$

157. If 4 (3x -2) = 2 (3x + 8), then x = ?

(a) 1 (b) 2
(c) 3 (d) 4

RRB NTPC 05.04.2016 Shift-1

Ans : (d) 4 (3x -2) = 2 (3x + 8)

12x – 8 = 6x + 16

12x – 6x = 16 + 8

6x = 24

$\Rightarrow x = \frac{24}{6} \Rightarrow x = 4$

158. Solve : if p=2 and q= –3 then $(p^2q - pq^2)$ = ...

(a) -6 (b) -30
(c) 6 (d) 30

RRB NTPC 04.04.2016 Shift : 2

Ans : (b) $p^2q - pq^2 = pq(p-q)$

$= 2\times(-3)[2-(-3)]$

$= -6(2+3)$

$= -30$

159. If $y = \frac{2x-1}{x+3}$ and y =1 then find the value of x.

(a) 4 (b) - 4
(c) 3/2 (d) 4/3

RRB NTPC 03.04.2016 Shift : 1

Ans : (a) $y = \frac{2x-1}{x+3}$

putting y =1

$1 = \frac{2x-1}{x+3} \quad \Rightarrow x+3 = 2x-1$

$\Rightarrow 2x - x = 3 + 1 \qquad \Rightarrow x = 4$

160. Find the value of x in 5x + 7y = 19, 7x + 5y = 17

(a) 1 (b) 2
(c) 3 (d) 4

RRB NTPC 03.04.2016 Shift : 3

Ans : (a) 5x + 7y = 19 ……..(i)
7x + 5y = 17 ……..(ii)

After subracting equation (i) × 5 equation (ii) × 7,

25x + 35 y = 95
49x + 35y = 119
– – – .
−24x = −24

$x = \frac{24}{24} = 1$

161. Solve: (3x + y) (2x – 3y)

(a) $6x^2 - 9xy - 3y^2$ (b) $6x^2 - 7xy + 3y^2$
(c) $6x^2 - 7xy - 3y^2$ (d) $6x^2 - 11xy + 3y^2$

RRB NTPC 31.03.2016 Shift : 1

Ans : (c) (3x+y) (2x – 3y)
$= 6x^2 - 9xy + 2xy - 3y^2$
$= 6x^2 - 7xy - 3y^2$

162. If x + 2y = 27 and x – 2y = – 1 then find the value of y.

(a) 3 (b) 4
(c) 7 (d) 6

RRB NTPC 31.03.2016 Shift : 2

Ans : (c) x + 2y = 27 …….(1)
x – 2y = –1 …….(2)

∴equation (1) − equation (2)

x + 2y = 27
x – 2y = –1
– + +
4y = 28
y = 7

163. Solve : (x + 2y) (2x –y)

(a) $2x^2 + 5xy + 2y^2$ (b) $2x^2 + 3xy - 2y^2$
(c) $x^2 + 4xy + y^2$ (d) $x^2 + 4xy - y^2$

RRB NTPC 29.03.2016 Shift : 3

Ans : (b) (x + 2y) (2x–y)
$= 2x^2 - xy + 4xy - 2y^2$
$= 2x^2 + 3xy - 2y^2$

164. P + Q = 2 (P – Q) if Q = 10, then find the value of P.

(a) 10 (b) 40
(c) 20 (d) 30

RRB NTPC 29.03.2016 Shift : 3

Ans : (d) P + Q = 2 (P–Q)
putting Q = 10,
P + 10 = 2 (P–10)
P + 10 = 2P–20
P = 30

165. If a+b = –2 and a–b = 12 then find the value of b?

(a) 5 (b) –5
(c) 7 (d) –7

RRB NTPC 28.03.2016 Shift : 2

Ans : (d) a + b = –2.....(i)
a – b = 12.....(ii)

adding, 2a = 10
a = 5

putting the value of a in equation (i)

5+ b = –2
b = –2 – 5
b = –7

166. If (2a/m + b/n) = 2 and (a/m-b/n) = 4 then find the value of 'a' and 'b' respectively.

(a) 2m, -2n (b) -2n, 2m
(c) 2m, 2n (d) -2m, 2n

RRB NTPC 18.01.2017 Shift : 1

Ans : (a) $\frac{2a}{m} + \frac{b}{n} = 2$.....(i)

$\frac{a}{m} - \frac{b}{n} = 4$.....(ii)

Adding equation (i) and (ii)

$\frac{2a}{m} + \frac{a}{m} = 6$

$\frac{3a}{m} = 6$

$\boxed{a = 2m}$

Again multiplying by 2 in equation (ii) then subtract of equation (i)

$\frac{2a}{m} - \frac{2b}{n} = 8$ .....(ii)×2

$\frac{2a}{m} + \frac{b}{n} = 2$ .....(i)

$-\frac{3b}{n} = 6$

$\boxed{b = -2n}$

So, a and b value is 2m, –2n respective y.

167. If $\left(x^2 + \frac{1}{16x^2}\right) = \frac{19}{2}$ then find the value of $\left(2x - \frac{1}{2x}\right)$

(a) 6 (b) 12
(c) 32 (d) 41

RRB NTPC 19.04.2016 Shift : 3

Ans : (a) $x^2 + \frac{1}{16x^2} = \frac{19}{2}$

$\Rightarrow 4x^2 + \frac{1}{4x^2} = \frac{19}{2} \times 4$ (multiply 4 in both side)

$\Rightarrow (2x)^2 + \frac{1}{(2x)^2} = 38$ .....(i)

$\because \quad \left(2x-\frac{1}{2x}\right)^2=(2x)^2+\frac{1}{(2x)^2}-2$

$=38-2=36$

$\Rightarrow 2x-\frac{1}{2x}=\sqrt{36}=6$

168. If 3.5x = 0.07y then find the result of [y-x/y+x]

(a) 51/49 (b) 49/53
(c) 49/51 (d) 53/57

RRB NTPC 18.04.2016 Shift : 1

Ans : (c) Given-

$3.5\,x = 0.07y$

$\frac{x}{y}=\frac{0.07}{3.5}=\frac{7}{350}$

$\frac{x}{y}=\frac{1}{50}$

$\therefore \frac{y-x}{y+x}=\frac{50-1}{50+1}=\frac{49}{51}$

169. If 0.08x + 0.04y = 10 and 0.2 (x–1) + 0.4y = 24.8 then find the value of x.

(a) 125 (b) 150
(c) 1.25 (d) 12.5

RRB NTPC 16.04.2016 Shift : 1

Ans : (a) From the question,

$0.08x + 0.04y = 10$

$\Rightarrow \frac{8}{100}x+\frac{4}{100}y=10$

$\Rightarrow 8x + 4y = 1000$

$\Rightarrow 2x + y = 250$ ……(i)

again

$0.2\,(x-1) + 0.4y = 24.8$

$\Rightarrow 2(x-1) + 4y = 248$

$\Rightarrow 2x - 2 + 4y = 248$

$\Rightarrow 2x + 4y = 250$

$\Rightarrow x + 2y = 125$ …….(ii)

Multiplying 2 in equation (i) then subtract to equation (ii)

$4x - x = 500-125$

$\Rightarrow 3x = 375$

$\boxed{x=125}$

170. Solve: x – 3 = 3x + 7

(a) 5 (b) –5
(c) 1 (d) $\frac{10}{4}$

RRB NTPC 11.04.2016 Shift : 1

Ans : (b) from the question,

$x - 3 = 3x + 7$

$\Rightarrow 3x - x = -3 - 7$

$\Rightarrow 2x = -10$

$\boxed{x=-5}$

171. Given w = – 2, x = 3, y = 0 and z = $-\frac{1}{2}$ then the value of $\frac{z}{w}+x$

(a) $3\frac{1}{4}$ (b) $-3\frac{1}{4}$
(c) 3.2 (d) 3.5

RRB NTPC 11.04.2016 Shift : 2

Ans : (a) w = - 2, x = 3, y = 0 and z = $-\frac{1}{2}$

$\frac{z}{w}+x=\frac{-\frac{1}{2}}{-2}+3$

$=\frac{1}{4}+3 \quad =\frac{13}{4}=3\frac{1}{4}$

172. Given w = –2, x = 3, y = 0 and z = $-\frac{1}{2}$ then find the value of $x\sqrt{(x+wz)}$

(a) ±6 (b) –6
(c) 6 (d) 5

RRB NTPC 07.04.2016 Shift : 3

Ans :(a) w = – 2 , x = 3,y = 0 & z = $-\frac{1}{2}$

$x\sqrt{(x+wz)}=3\sqrt{\left(3+(-2\times\frac{-1}{2}\right)}$

$=3\sqrt{(3+1)}=3\sqrt{4}=3\times(\pm2)=\pm6$

173. If 22x – 40 = 207 + 3x, then x = ?

(a) 14 (b) 13
(c) 12 (d) 11

RRB NTPC 06.04.2016 Shift : 2

Ans : (b) $22x - 40 = 207 + 3x$

$22x - 3x = 207 + 40$

$19x = 247$

$x=\frac{247}{19}$,

$\therefore x = 13$

Type - 11

174. If $(a^2-b^2)\div(a+b)=25$ then the value of a-b is-

(a) 15 (b) 18
(c) 25 (d) 30

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (c) Given- $(a^2-b^2)\div(a+b)=25$

$\frac{(a-b)(a+b)}{(a+b)}=25$

$\boxed{a-b=25}$

175. If $x + y = 1$, then $x^3 + y^3 + 3xy - 1 = ?$

(a) 0 (b) 1
(c) 5 (d) 2

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (a) $x + y = 1$
Cube of both side
$(x+y)^3 = (1)^3$
$x^3 + y^3 + 3xy\,(x+y) = 1$ [$\because x + y = 1$]
$x^3 + y^3 + 3xy - 1 = 0$

176. If $a^2 + b^2 = 80$ and $a - b = 4$, then ab = ?

(a) 20 (b) 24
(c) 28 (d) 32

RRB NTPC 05.04.2016 Shift-1

Ans : (d) Given-
$a^2 + b^2 = 80$, $a - b = 4$
$\because (a - b)^2 = a^2 + b^2 - 2ab$
$\Rightarrow (4)^2 = 80 - 2ab$
$\Rightarrow 2ab = 80 - 16$
$\Rightarrow 2ab = 64$
$\Rightarrow ab = \frac{64}{2} \Rightarrow ab = 32$

177. Find $(x+y)^2 - (x-y)^2 = ?$

(a) $2x^2y^2$ (b) $4xy$
(c) $2x^2 + 2y^2$ (d) $x^2 - y^2 + 2xy$

RRB NTPC 03.04.2016 Shift : 2

Ans : (b)
$(x+y)^2 - (x-y)^2 = (x^2 + y^2 + 2xy) - (x^2 + y^2 - 2xy)$
$= x^2 + y^2 + 2xy - x^2 - y^2 + 2xy$
$= 4xy$

178. $(a-b)^2 + 2ab = ?$

(a) $a^2 - b^2$ (b) $a^2 + b^2$
(c) $a^2 - 4ab + b^2$ (d) $a^2 - 2ab + b^2$

RRB NTPC 30.03.2016 Shift : 1

Ans : (b) $(a-b)^2 + 2ab$
$= a^2 + b^2 - 2ab + 2ab$
$= a^2 + b^2$

179. Expand: $(c-3)^3$

(a) $c^3+9c^2+27c+27$ (b) $c^3-9c^2+27c-9$
(c) $c^3-9c^2+27c-27$ (d) $c^3-9c^2-27c-27$

RRB NTPC 30.03.2016 Shift : 2

Ans : (c) $(c-3)^3 = c^3 - 27 - 9c^2 + 27c$
$= c^3 - 9c^2 + 27c - 27$
[$\because (a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$]

180. Expand : $(s + 2)^3$

(a) $s^3 + 3s^2 + 12s + 8$ (b) $s^3 + 3s^2 + 6s + 8$
(c) $s^3 + 6s^2 + 12s + 8$ (d) $s^3 + 6s^2 + 6s + 8$

RRB NTPC 29.03.2016 Shift : 3

Ans : (c) $(s + 2)^3 = s^3 + (2)^3 + 3 \times s \times 2\,(s + 2)$
$= s^3 + 8 + 6s^2 + 12s$
$= s^3 + 6s^2 + 12s + 8$

181. Given $(a^2 + b^2) = 60$, then find the value of $(a + b)^2 + (a-b)^2$

(a) 90 (b) 120
(c) 140 (d) 150

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) Given–
$a^2 + b^2 = 60$ ……. (i)
$\therefore (a + b)^2 + (a-b)^2 = (a^2 + b^2 + 2ab) + (a^2 + b^2 - 2ab)$
$= a^2 + b^2 + 2ab + a^2 + b^2 - 2ab = 2(a^2 + b^2)$
$= 2 \times 60$ from equation (i)
$= \boxed{120}$

182. If $40x^2 = 334^2 - 134^2$ then value of x^2 is–

(a) 2340 (b) 234
(c) 1234 (d) 144

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) From the question,
$40x^2 = 334^2 - 134^2$ $\{a^2 - b^2 = (a+b)(a-b)\}$
$\Rightarrow 40x^2 = (334 - 134)(334+134)$
$\Rightarrow 40x^2 = 200 \times 468$
$\Rightarrow x^2 = 2340$

183. If $a - b = 5$ and $a^2 + b^2 = 97$ then ab = ?

(a) 48 (b) 32
(c) 36 (d) 72

RRB NTPC 27.04.2016 Shift : 2

Ans : (c) Given-
$a - b = 5$, $a^2 + b^2 = 97$
$(a-b)^2 = (5)^2$
$a^2 + b^2 - 2ab = 25$ ....(i)
take both side square
putting value of a^2+b^2 in eq^n (i)
$97 - 2ab = 25$
$-2ab = -72$
$ab = 36$

184. $(4x - 7)^2 = ?$

(a) $4x^2 - 56x + 49$ (b) $4x^2 - 14x + 49$
(c) $16x^2 - 14x + 49$ (d) $16x^2 - 56x + 49$

RRB NTPC 27.04.2016 Shift : 2

Ans : (d)
$\because (a - b)^2 = a^2 - 2ab + b^2$
$\therefore (4x-7)^2 = 16x^2 - 56x + 49$

185. If a-b = 5 and $a^2+b^2 = 31$ then find the value of ab –

(a) 4 (b) 5 (c) 2 (d) 3

RRB NTPC 28.04.2016 Shift : 1

Ans : (d) Given $a - b = 5$, $a^2 + b^2 = 31$
$(a - b)^2 = a^2 + b^2 - 2ab$
$5^2 = 31 - 2ab$
$25 = 31 - 2ab$
$2ab = 31 - 25 = 6$
$\boxed{ab = 3}$